

Annual Health Report Series

**Improving Maternal, Newborn
and Child Health Outcomes
in a Decade**

Pakistan Report 2020



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ACRONYMS

AARR	Average Annual Rate of Reduction	MPDSR	Maternal Perinatal Death Surveillance Response
ANC	Antenatal Care	MSDs	Medical Store Depots
ARI	Acute Repertory Illness	NATPOW	National Trust for Population Welfare
BHU	Basic Health Unit	NCDs	Non Communicable disease
CCI	The Council of Common Interests	NMR	Neonatal Mortality Rate
CPR	Contraceptive Prevalence Rate	PDHS	Pakistan Demographic Health Survey
CRVS	Civil Registration and Vital Statistics	PHC	Primary Health Care
DHIS	District Health Information System	PMMS	Pakistan Maternal Mortality Survey
DHQ	District Head Quarter	PMs	Permanent Methods
ENPA	Every Newborn Action Plan	PNC	Postnatal Care
EPI	Expanded Programme for Immunization	PSBI	Possible Suspected Bacterial infections
FALAH	Family advancement of life and health	PSI	Population services International
FATA	Federally Administered Tribal Area	PTF	Provincial Population Task Forces
FP	Family planning	RHC	Rural Health Centre
FPAF	Family Planning Association of Pakistan	RMNCH	Reproductive Maternal Neonatal Child and Adolescent Health
ICT	Islamabad Capital Territory	SAAVY	Sample Vital Statistics and Verbal Autopsy
IMCI	Integrated Management of Childhood illnesses	SARA	Service Availability and Readiness Assessment
IMR	Infant Mortality Rate	SBA	Skilled Birth Attendant
KMC	Kangaroo Mother Care	SDGs	Sustainable Development Goals
LARCs	Long Acting Reversible Contraceptives	SNL	Saving Newborn Lives
LHV	Lady Health Visitor	THQ	Tehsil Head Quarter
LHWs	Lady Health Workers	UNFPA	United Nation Population Fund
MMR	Maternal Mortality Rate	WMO	Women Medical Officer

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EXECUTIVE SUMMARY

Pakistan has committed to Sustainable Development Goals and targets to be achieved by 2030. These include reducing preventable maternal, neonatal and child mortality and attaining universal health coverage. Ensuring reproductive health rights and provision of quality family planning services remain central to these efforts. With exactly a decade to reach the 2030 mark, this report endeavours to review the past progress and identifies key areas for future action.

The Government of Pakistan has recently endorsed the evidence-based Basic Package of UHC interventions at the district level based on the directions inferred from the WHO document on Disease Control Priorities (Third Edition). The Global Financing Facility hosted at World Bank has included Pakistan in the list of eligible countries and an investment case is being prepared in this regard. Consultations are underway to identify the required health system reforms and re-imagine the district health and social protection systems to effectively deliver this Basic Package of interventions.

It is indeed timely to synthesize existing data to understand the lapses in the health system and highlight issues of measurements and monitoring. This report analyses gaps and barriers for effective quality coverage of family planning and maternal, newborn and child health (MNCH) services. The topics of nutrition and adolescent health, though intricately linked with MNCH outcomes, will be covered in detail in the next annual report.

Methodology

Programmatic interventions for reducing preventable maternal and child mortality and strengthening family planning are dependent upon the ability of the health system to deliver equitable and quality health services. Measuring coverage of key Family Planning and MNCH coverage indicators has been the usual practice. There is a growing realization globally that the increasing rates of skilled birth attendance and facility-based deliveries have not translated into equivalent health and survival outcomes for mothers and newborns; and thus, measuring quality of care and outcomes is equally important. In addition, the service coverage needs to be understood in terms of a client's interface with the health system. A systematic framework is thus necessary to measure dimensions of lack of access, gaps in service readiness and effective coverage, lapses in the provision of quality care and user factors that further reduce the potential impact of interventions on specific RMNCH outcomes. This report uses the cascade framework¹ in an effort to capture a holistic picture.

Relevant and accessible data sources were synthesized for the assessment of progress in recent years, including national and subnational surveys, facility surveys, routine health facility data, health system data and research findings.

¹ Amouzou A, et al. BMJ Glob Health 2019;4:i114–i124. doi:10.1136/bmjgh-2018-001297

Main Findings

Since 1990, Pakistan has steadily reduced maternal, newborn and under-five mortality rates and improved use of modern methods of family planning. The current average annual rates of improvement need significant acceleration to meet the SDG targets for health by 2030. The review of existing data has highlighted multiple losses in the cascade to effective coverage from availability, accessibility and quality of services to the actual attainment of health impact.

During the last few decades, fertility has been declining in Pakistan at a slow pace but slowed down in the most recent period 2013-2018 (average annual rate of decline of 1.1% per year), despite the stronger political commitment to reducing fertility. Total fertility was 3.6 children per woman by 2017/18, fueling a continued rapid population growth and a relatively large proportion of higher risk deliveries. . The use of modern contraceptives in Pakistan remained low (25.0% of currently married women in PDHS 2017/18), and the desire for children high (3.9 children for women, 4.3 for men). If couples want fewer children in the near future, the demand for modern contraceptives will increase rapidly, and further strengthening of the family planning (FP) services is needed. Multiple interventions need to be strengthened to increase FP coverage in line with government targets such as promoting post-partum FP to avail missed opportunity of rising institutional deliveries, addressing the causes of contraceptive stock outs and scaling up safe post abortion care services.

The 2019 maternal mortality survey shows a decline in maternal mortality ratio (MMR) to 186 per 100,000 live births, from 276 in the PDHS 2006/07. This corresponds with an average annual rate of reduction of 3.1% per year between 2005 and 2018². To reach the 2030 target of 70 per 100,000 live births more than a doubling of the rate of decrease in the last decade (average annual rate of decline of 8.3%) is required. Progress was made on several indicators of antenatal delivery care: ANC coverage increased to 84% (PLSMS 2018/19), more than half of pregnant women made their first visit in the first trimester, half of women completed at least four visits (PDHS 2017/18) and institutional deliveries rose to 70%. About 60% occurred in private facilities according to PSLM, but the share of the public sector is growing with an escalation in number of primary health care facilities operational 24/7. Quality of care is a concern and respectful maternity care needs more emphasis. Unnecessary caesarean deliveries are rising amongst the richest and highly educated women. Data gaps from private sector services and on causes and drivers of maternal deaths are noted. Areas of programmatic improvement include strengthening CRVS, maternal death notification and expanding MPDSR to all districts.

Newborn mortality rate has fallen to 41 per 1,000 live births and still birth rate to 31 per 1,000 total births. Both neonatal mortality and stillbirth rates decreased slowly at an annual rate of reduction of only 2% in the last decade. Research showed major gaps in the quality of neonatal interventions. To fasttrack progress, attention is needed to ensure quality of care around the time of birth and to invest in small and sick newborn care. Perinatal death reviews are a must to review the case management and make

² 2005 and 2018 are the midpoints of the 3-year period of the retrospective maternal mortality estimates.

incremental improvements. A national neonatal survival strategy with provincial benchmarks, costed plan and adequate financing is critical to accelerate progress. Human Resource strategy including task shifting and competence based capacity building is needed. A plan for developing cadres of neonatologists, neonatal nurses and auxiliaries for neonatal intensive care units is urgently required.

Under-five mortality has dropped steadily during the last two decades at an annual rate of decline just under 3% to 67 per 1,000 live births by 2019. Sixty percent of all under-five deaths are due to neonatal causes. Hence a major improvement in neonatal survival will be essential for reaching under-five mortality rate 2030 target. Pneumonia and diarrhea are the leading causes of post neonatal and early child mortality. Access and utilization of services continued to improve for ARI and childhood diarrhoea. 70-80% of sick under-fives were taken to a health facility, and 53% of those with diarrhoea received ORS fluids in 2018/19. Coverage of fully immunized children increased to 66% in 2017/18 and further increases in pentavalent third dose and measles vaccination were observed in the PSLM 2018/19. Better targeting is necessary: urban-rural and provincial inequalities persist and the poorest wealth quintile has much lower coverage. Quality of pediatric care is a challenge whereby nutritional assessment, age-appropriate vaccination, ORS plus zinc and continued feeding during diarrhea and provision of antibiotic for managing ARI could be improved by strengthening knowledge and counseling skills of service providers. All interventions for under-fives should pay special attention to the children from the poorest households which still have lower coverage and higher mortality.

Continuity of essential health services is indispensable to safeguard the modest gains made during the last decades. Preparedness is compulsory to withstand the effect of natural disasters and health emergencies. The COVID-19 pandemic hammered down this necessity once again. Family planning and MNCH services were disrupted with the highest drop in out-patient consultations and deliveries during the period April-May 2020 with strict lock down enforcement. However, a quick rebound was noted in June and thereafter. All reproductive health programming needs to be risk-informed and should be attentive to build resilience of health system in face of any future health crises.

Call to Action

Pakistan has made several positive strides in improving maternal, newborn and child health of its people which should form the basis for a much greater acceleration of progress in the coming decade to reach the 2030 target. Wider health service coverage must be matched by concomitant improvements in quality of care if maternal, neonatal and under-five mortality is to be reduced substantially. Challenges of health system, equity and high unmet need for family planning need urgent attention to accelerate progress in the next decade. This calls for fundamental health sector and non-health sector actions for delivering a priority RMNCH package of quality services to the women and children who need it most.

Accelerating the progress on Family Planning, Maternal, Newborn and Child Health in a decade demands a de novo approach, high political commitment backed by adequate financing and the will to make difficult decisions for addressing redundancies and gaining efficiency. The recent developments on RMNCAH+N investment case, UHC Basic Package and renewed government commitment to invest in

health have given an impetus to think bolder and bigger. Prioritizing action to reach the SDG and related targets by 2030 is urgently required. Focusing on meeting the unmet need of FP, using the opportunity of high institutional births for PPFP ; identifying the most un-served populations, improving the management of the top causes of mortality in both public and private sectors e.g. PPH, eclampsia, sepsis, prematurity, birth asphyxia, diarrhoea and pneumonia can give the highest reduction in burden of deaths.

The ten priorities for action in ten years (10X10 agenda) are

1. Improving RMNCH leadership and coordination mechanisms by strengthening TWGs at national and provincial levels
2. Developing RMNCH acceleration package including family planning for immediate scale-up
3. Providing operational support to RMNCH acceleration package by UHC structures or RMNCH+N directorates with quality of care units to provide support for implementing evidence-based MNCH and FP standards
4. Harnessing the role of a strong private sector, already catering to two-thirds of the RMNCH needs. Partnerships with professional associations and health care commissions can provide leverage in this regard.
5. Targeting the poor, the rural and the un-educated is imperative for reducing the coverage disparities. Linking health sector efforts with poverty reduction, nutrition sensitive and social protection initiatives will bring efficiency and reduce duplication.
6. Resolving the serious health work force crises, nurse midwife ratio to population and attaining sufficient numbers of neonatologists and neonatal nurses.
7. Improving timely referral of complicated cases by expanding pooled ambulance services and linking district level facilities with pre-identified teaching hospitals. Laying out clear referral protocols and responsibilities at each level of health system is needed.
8. Digital innovations for improving data recording and reporting as well as improving LMIS and increasing awareness through mobile messaging
9. Investing in research and information system for tracking effective quality coverage; accounting for the private sector; bridging data gaps for CRVS (Civil Registration and Vital Statistics); birth weight measurement, reporting of still births and standardized measures for determining quality of care and cohort tracking. Extensive investment is required to scale-up and streamline MPDSR and link these efforts with continuous quality improvement.
10. Developing inter-sectoral linkages for addressing the underlying risk factors of low female social status, malnutrition, air pollution and inadequate WASH behaviours.

INTRODUCTION

Background

Pakistan is committed to the Sustainable Development Goals and national targets have been set for the health sector. With a decade to achieve 2030 targets, the National Ministry of Health Services Coordination and Regulation and Health Services Academy are initiating a series of annual reports to review progress and guide action. This first report focuses on family planning, maternal, newborn and child health; and reviews the progress on SDGs and related targets for MNCH. The objective is to inform the Parliamentary Committee for Health and stakeholders for political advocacy and to garner the required policy and budgetary support. The timing of the preparation of this report has coincided with current dialogue on development of an investment case for Universal Health Care Benefit Package and planning of RMNCAH+N through a trust fund by the Global Financing Facility to Pakistan. This report has benefitted from and intends to contribute to these simultaneously on-going processes by highlighting trends, inequalities in access and utilization of RMNCAH+N services and identifying priorities for action.

This work has been possible through the generous grant of Bill and Melinda Gates Foundation to Countdown to 2030; to improve coverage measurement and monitoring of RMNCAH+N implementation, and strengthen the regional and country capacity for evidence generation and use. Countdown to 2030 provides independent monitoring and rigorous analysis that are crucial to accountability. This Annual Review Report on Pakistan's Progress for SDG and related targets for RMNCH 2020 is a collaborative effort by Health Services Academy, Ministry of Health Services Coordination and Regulation, the National Ministry for Health Services Coordination and Regulation and the University of Manitoba.

Pakistan has displayed commendable commitment to the 2030 Agenda for Sustainable Development as it was one of the first countries to endorse it globally in 2015. On 16th February 2016, the Parliament unanimously approved the Sustainable Development Goals (SDGs) as the national development agenda. It also formed the Parliamentary SDGs Secretariat based at the National Assembly – one of the few countries to do so. This process of legislation was the first and crucial step in mainstreaming and localizing the SDGs.

Subsequently, the government of Pakistan has held discussions with all stakeholders for coordinating and strengthening efforts at federal and provincial levels to achieve Pakistan's sustainable development and poverty reduction targets. The consultation process emphasized the need for national categorisation of SDGs, improved data collection and enforcement of monitoring mechanisms.

A dedicated SDGs Section³ has been formed at the federal level in the Ministry of Planning, Development & Special Initiatives (MoP&SI) to monitor and coordinate as a national coordinating entity. The efforts are

³ <https://www.sdgpakistan.pk/web/sdgs>

aimed to accomplish the following outcomes:

1. Main streaming SDGs in plans, policies and resource allocation aligned to 2030 Agenda
2. SDGs monitoring, reporting and evaluation capacities strengthened
3. Financing flows increasingly aligned with 2030 Agenda
4. Innovative approaches applied to accelerate progress on priority SDGs

The inter-ministerial efforts are also replicated at sectoral levels. In 2019, a SDG localization report⁴ on Health was prepared by Health Services Academy, Islamabad in partnership with MoHSR&C, provincial/regional health departments and WHO. The report called for filling the existing gaps in information, and for developing a regular system of reporting and monitoring progress in health with respect to SDGs.

Current Situation of Family Planning and MNCH in Pakistan

Pakistan, as the fifth most populous country in the world has its fair share of health challenges. Health system inadequacies, poverty, malnutrition, high fertility rate, low social status of women, environmental problems and risk of natural disasters are some of the realities hindering the achievement of SDG targets. The country lags well behind most developing countries in access to essential health services and health equity.

Table 1: Pakistan vis-a-vis its regional neighbours

	Pakistan	Afghanistan	Bangladesh	Bhutan	India	Nepal	Sri Lanka
CPR	27.6	24.9	57.5	63.9	50.3	47.8	54.7
MMR	140	638	173	183	145	186	36
NMR	41	37.1	17.1	16.4	22.7	19.9	4.5
U5MR	69	62	30	30	37	32	7

Pakistan is second to Sri Lanka in Maternal Mortality Ratio but has fallen behind its neighbours in contraceptive prevalence rates and neonatal and under-five mortality.

The country's extremely high fertility is the result of inadequate and fragmented services that fail to meet contraceptive needs and a preference for large families that reflects women's traditional status. Neonatal mortality is among the highest in the world and half of the infant deaths result from poor maternal health and nutrition. High out of pocket health expenditures and social constraints present serious constraints to use limited services. However, some positive developments over the last few decades resulting in the reduction of maternal mortality, improvement in skilled birth attendance and immunization coverage, have been noted⁵.

The current population growth rate is estimated around 2.1 % in 2018⁶. Pakistan still has a high fertility

⁴ Saima Hameed, Mariyam Sarfraz 2019 SDG-3 Localization in Pakistan, Health Services Academy Islamabad, MoHSR&C and WHO.

⁵ Jamil Arshad and Raza Zaidi, Situation Analysis 2020 Universal Health Coverage PAKISTAN:

Towards Health for All by All in Pakistan, MOHSR&C, Provincial Governments of Punjab, Sindh, KPK, Baluchistan, GB and WHO.

⁶ <https://data.worldbank.org/indicator/SP.DYN.CDRT.IN?locations=PK>

rate of 3.6 children per woman in PDHS 2017-18, with higher levels in rural areas. Life expectancy at birth has improved to 68 years (66 years for males and 70.1 years for females) in 2017 and is slightly lower than the global average⁷. Maternal, newborn and child mortality constitute the largest share of preventable deaths in Pakistan. There are an estimated 4.7 million births annually with more than 8,000 maternal deaths, 248,000 newborn deaths, 184,000 still births and 405,000 under-five deaths. About 3,500 obstetric fistula cases are reported every year. The Contraceptive Prevalence Rate has stagnated recently despite the high unmet need for family planning. About 2.5 million abortions are estimated on an annual basis. The COVID-19 pandemic has also adversely affected the utilization of essential MNCH services and any future plans should consider the ramifications.

Pakistan has recently finalized a basic package of Universal Health Care for implementation at the district level. This package is based on evidence based priority interventions selected from the WHO document on Disease Control Priorities (3rd Edition). The prioritization of interventions was based on demographic needs, epidemiological trends, burden of diseases and cost effective-ness of individual interventions. Reproductive, maternal, newborn and child health interventions are featured prominently in this package.

Methods

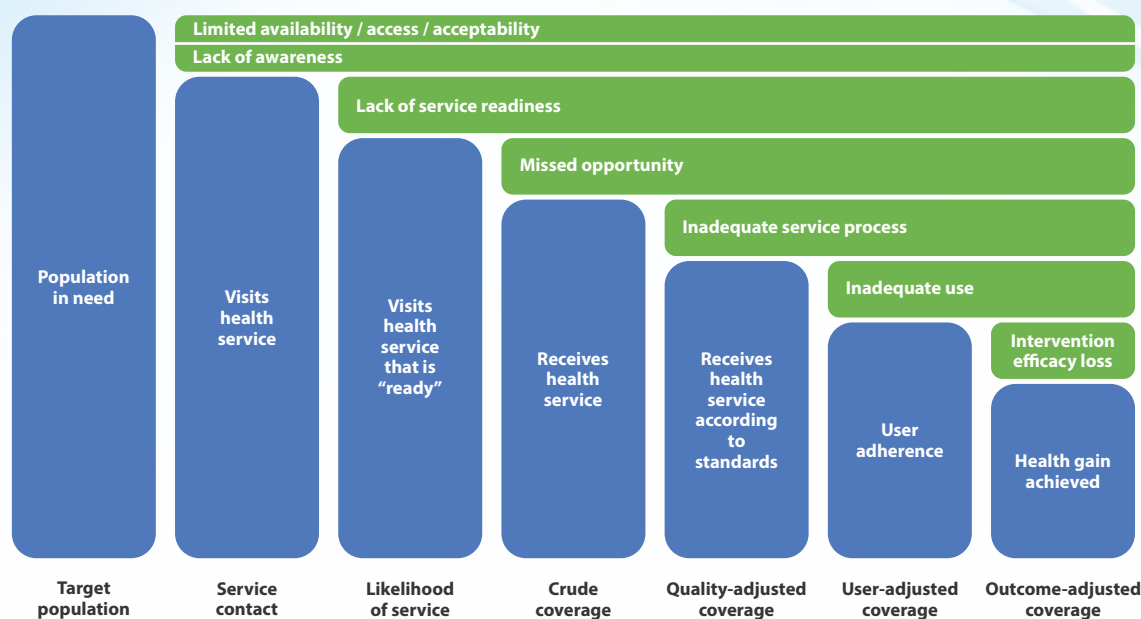
This report uses the conceptual framework of measuring availability, accessibility, affordability, quality and effective coverage of key interventions. Disparities in coverage are highlighted and DHIS data from public health facilities is explored for disruptions and trends in the continuity of RMNCH services. Each section highlights the policies and programmes being implemented, reviews the rates of progress since 1990 and projects future trends. The conclusion section presents a set of scenarios and impact calculations using the Lives Saved Tool (LiST).

Triangulation of different sources of information is employed to assess health system gaps and identify priorities for action. In an effort to identify where the main gaps are, a conceptual framework of the cascade of care is used that integrates dimensions of lack of access, gaps in service readiness, lapses in the provision of quality care and user factors that further reduce the potential impact of interventions on specific RMNCH outcomes⁸.

⁷ Institute of Health Metrics & Evaluation (IHME), 1990-2017; BOD data for Pakistan: <https://vizhub.healthdata.org/gbd-compare/>

⁸ Amouzou A, et al. BMJ Glob Health 2019;4:i114–i124. doi:10.1136/bmjgh-2018-001297

1. The Conceptual Framework



This report uses data from household surveys (PDHS 2017-18 and PSLM 2019) and information from public health facilities (health facility assessments and DHIS). Findings from recent MICS Surveys are also included. Literature review of recent journal articles and grey literature has been used to fill in the information gaps. The relevant government policies and programmes are presented in each chapter to highlight the efforts to improve service delivery and monitoring of progress. Lessons derived from these endeavours are important to shape the future recommendations especially to bridge the hiatus between service contacts and content and quality of services received (WHO standards followed, user adherence and patient satisfaction).

The chapters on family planning, maternal health, newborn health and child health highlight the fault lines in the health system service delivery and propose key actions to drive progress towards SDGs. The last chapter pulls out the common gaps and priority action areas in the coming decade. It emphasizes the need to invest in measurements to better monitor, understand and act on the gaps in effective coverage. Pakistan has a multitude of regular household surveys but reliable real time monitoring needs more investment in DHIS, MPDSR and sentinel surveillance for maternal, newborn and child deaths. Making significant progress towards universal health coverage with quality services for women's and children's health is possible with focused planning, regular oversight and monitoring to make course corrections on the way.

CHAPTER 1. FAMILY PLANNING / REPRODUCTIVE HEALTH

Table 2: Main points

	Progress	Challenges
Policy / context	Government commitment to making FP accessible to all: National Council of Common Interests recommendations to achieve SDGs have been translated into a Draft Action Plan (2019-24) approved by the Federal Task Force, with specific FP targets, and executed by Provincial Task Forces	Women and men in Pakistan want almost twice as many children as the government target for 2030 (PDHS 2017/8)
Fertility	Total fertility declined to 3.6 children per woman by 2017/18, at an average annual rate of decline of 1.1% per year	The fertility decline is decelerating and was the slowest in three decades during 2012/13-2017/18
Coverage	Recent data from a FP 2020 self-report survey suggest an increase in contraceptive use	FP coverage (demand satisfied with modern methods) was low and not increasing by PDHS 2017/18;
Service readiness	Engagement and collaboration between public and private sector in supplying FP services	Stock-outs in public facilities were common in 2019/2020
Quality of care	No data	Almost no data are available; abortion rates are high
Covid-19	Covid-19 impacted the health services but a rebound in the 2 nd half of 2020 is likely	Major impact of Covid-19 pandemic on the distribution of contraceptives to women during Apr-Jun 2020, according to health facility data
Way forward	If couples want fewer children in the near future, the demand for modern contraceptive will increase rapidly, and further strengthening of the FP services is needed. Multiple interventions need to be strengthened to increase FP coverage in line with government targets such as promoting post-partum FP to avoid missed opportunity of rising institutional deliveries, posting of counselors of Population Welfare outlets at high volume facilities for post-partum FP counseling and all mothers are provided pre-discharge FP counseling and services; using DHIS dash-boards to monitor stock-outs and trends and employ SMS and other digital technologies for timely alerts and updates; streamlining procurement and logistics management of contraceptives; expanding FP outlets by Public Private Partnerships with GP clinics and pharmacies; exploring online supply of condoms, oral pills and injectables for already registered FP users and targeting men for increasing use of FP, given their role in couple's decision-making for adopting a method.	

Introduction

This chapter briefly reviews the main policies and targets for family planning and population, describing the main program implementation efforts and highlighting the larger epidemiological context of population fertility trends in Pakistan.

The assessment of progress uses the cascade of effective coverage. Several steps in the cascade have limited data but based on the national surveys (especially PDHS 2017/18), health facility data and research it is possible to identify areas of progress and major challenges. This chapter assesses:

- level and trends of the demand for children / family planning,
- knowledge and access to services, readiness and quality of services,
- extent to which services are received and used (contraceptive prevalence) and
- effective coverage of FP (demand for modern methods satisfied).

Main Policies

By 2030, the SDG target (3.7.1) is that all women of reproductive age should have access to modern family planning methods. In 2017, the Government of Pakistan made a commitment at the Family Planning 2020 Summit, which aimed to reach all citizens with high quality FP information and services⁹. This included raising the contraceptive prevalence rate to 50% by ensuring the optimal involvement of the public and private population and health sectors in family planning and move towards universal coverage of reproductive health services meeting the SDG target 3.7 by 2030; offering greater contraceptive choices through an improved method mix; halving unmet need among women and girls by 2020 and half of all married men knowing where birth spacing services can be obtained; reducing the unmet needs among married adolescent girls; and increased funding for FP in provincial and federal budgets.

The National Council of Common Interests has made recommendations to achieve SDGs related to population and health. The recommendations of Council have been translated into a Draft Action Plan (2019-24) approved by the Federal Task Force under the Chair of the President of Pakistan. The action plan helps to evaluate the extent of progress towards international commitments to achieve SDG targets related to population and health. The plan for implementation of Council's recommendations¹⁰ aims to:

- increase the present Contraceptive Prevalence Rate of 34% to 50% by 2025 and 60% by 2030;
- reduce the present Total Fertility Rate 3.3 births per woman to 2.8 births by 2025 and 2.2 births per woman by 2030;
- lower the present average population growth rate of 2.4% to 1.5% by 2025, and 1.1% by 2030.

⁹ http://www.familyplanning2020.org/sites/default/files/Govt_Pakistan_FP2020_Commitment_2017_0.pdf

¹⁰ Pakistan Economic Survey Report 2019-2020

Program implementation efforts

Governmental efforts have included the launch of family planning programmes in the 1960s through a network of provincial population welfare departments. A bigger push for providing contraceptive counseling and services at the doorstep of rural women came through induction of LHWs. Private sector contributions were significant through social marketing ventures like Green Star by PSI. In addition to public sector outlets, national and international NGOs have been also involved e.g. Family Planning Association of Pakistan (FPAP), Marie Stopes International and a consortium of NGOs called NATPOW. Population Council, UN agencies and bilateral donors like USAID and DFID have also supported different programmes on ground. For example, a major initiative FALAH was implemented from 2007-2012 in 20 districts and reported increasing contraceptive use by 8% reducing the unmet need for birth spacing from 14.2% to 10.8% over the same period¹¹. Post-partum family planning remains a window of opportunity in scenario of increasing facility births¹². Postpartum intrauterine contraceptive device was found to be an effective, acceptable contraception with fewer complications for the patients¹³.

Provinces have prepared multi-sectoral plans and allocated major funds for engaging other sectors including the private sector, Provincial line Departments, etc. These efforts aim to enhance the quality and coverage of services and ensure commodity security. Meetings of the Provincial Population Task Forces (PTF) (constituted in pursuance of the recommendations approved by the Council of Common Interest (CCI), are being held under the duress of the respective Chief Ministers to carry forward the agenda. Most provinces have included Family Planning commodities in their essential drugs list. Contraceptive procurement is a challenge for some provinces, because taking part in international competitive bidding with small quantities is not feasible. Other issues include integration of family planning services at facilities of Department of Health and involvement of Lady Health Workers in the family planning program in some provinces; access barriers; lack of quality services; religious, cultural and social norms; inadequate financing and lack of field staff due to long term ban on recruitments by the courts in Sindh and Baluchistan.

General epidemiological trends: slowdown of fertility decline

The total fertility rate during the 1970s and 1980s remained above six births per woman¹⁴. Data from national surveys shows a steady decrease in total fertility rates since 1990-91 (Figure 1). There was however a deceleration of the speed of the fertility decline to 1.1% average annual reduction rate (AARR) during the two most recent surveys.

¹¹ <https://www.popcouncil.org/research/family-advancement-for-life-and-health-falah>

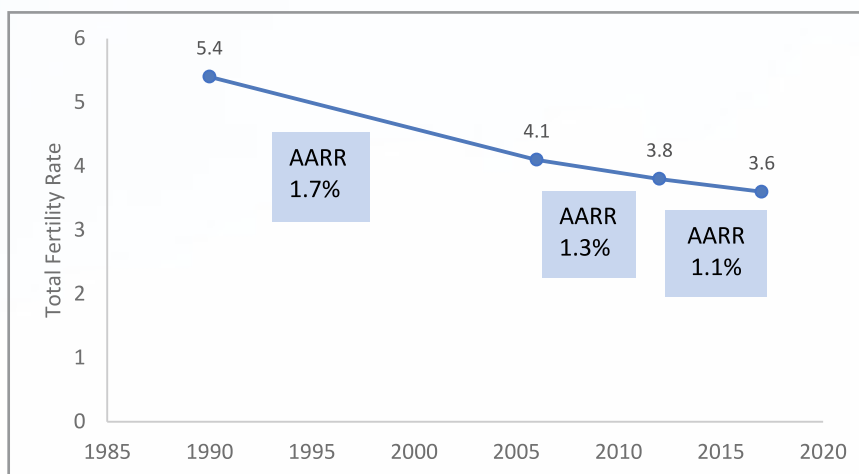
¹² Abbasi Y, Shaikh SR, Memon KN. Barriers and missed opportunities towards immediate and early post-partum family planning methods in Pakistan. *Professional Med J* 2020; 27(7):1448-1453.

¹³ Wasim T, Shaukat S, Javed L, Mukhtar S. Outcome of immediate postpartum insertion of intrauterine contraceptive device: Experience at tertiary care hospital. *JPMA the Journal of the Pakistan Medical Association*. 2018; 68(4):519-25.

¹⁴ Sathar, Z. A., & Casterline, J. B. (1998). The onset of fertility transition in Pakistan. *Population and development review*, 773-796.

Figure 1: Trends of total fertility rate (children per woman), Pakistan DHS surveys (AARR = Average Annual Rate of Reduction)

2. Trends of total fertility rate (children per woman), Pakistan DHS surveys (AARR = Average Annual Rate of Reduction)



Between PDHS 2012-13 and 2017-18, the median age at first marriage increased from 19.5 years to 20.4 years among women age 25-49. Similarly, the median age at the time of the first marriage increased from 24.7 years to 25.9 years among men age 30-49. During the same period, the percentage of women who were married by age 18 declined from 35% to 29%, while the percentage of men who were married by age 22 declined from 31% to 24%. The median age at first birth increased to 22.8 years, as compared with 22.2 years in 2012-13 (women age 25-49). About one in 12 teenage women age 15-19 had begun childbearing which was almost twice as high among young women with no education having started childbearing.

Assessment of progress

Desire for children: high ideal family size persistent

With ambitious goals to slow down population growth and reduce total fertility rates, the desire for children is an important indicator. In general, individual's fertility aspirations were much higher than Governmental targets. The mean ideal family size among women 15-49 in the PDHS 2017/18 was 3.9 children, which was only modestly lower than the 2012/13 survey (4.1 children)¹⁵. Women in the poorest quintiles (5.0 children), with no education (4.5 children) and rural women (4.2 children) have the higher ideal family sizes than their respective counterparts. Men's ideal family size was 4.3 children, which was the same as in 2012/13.

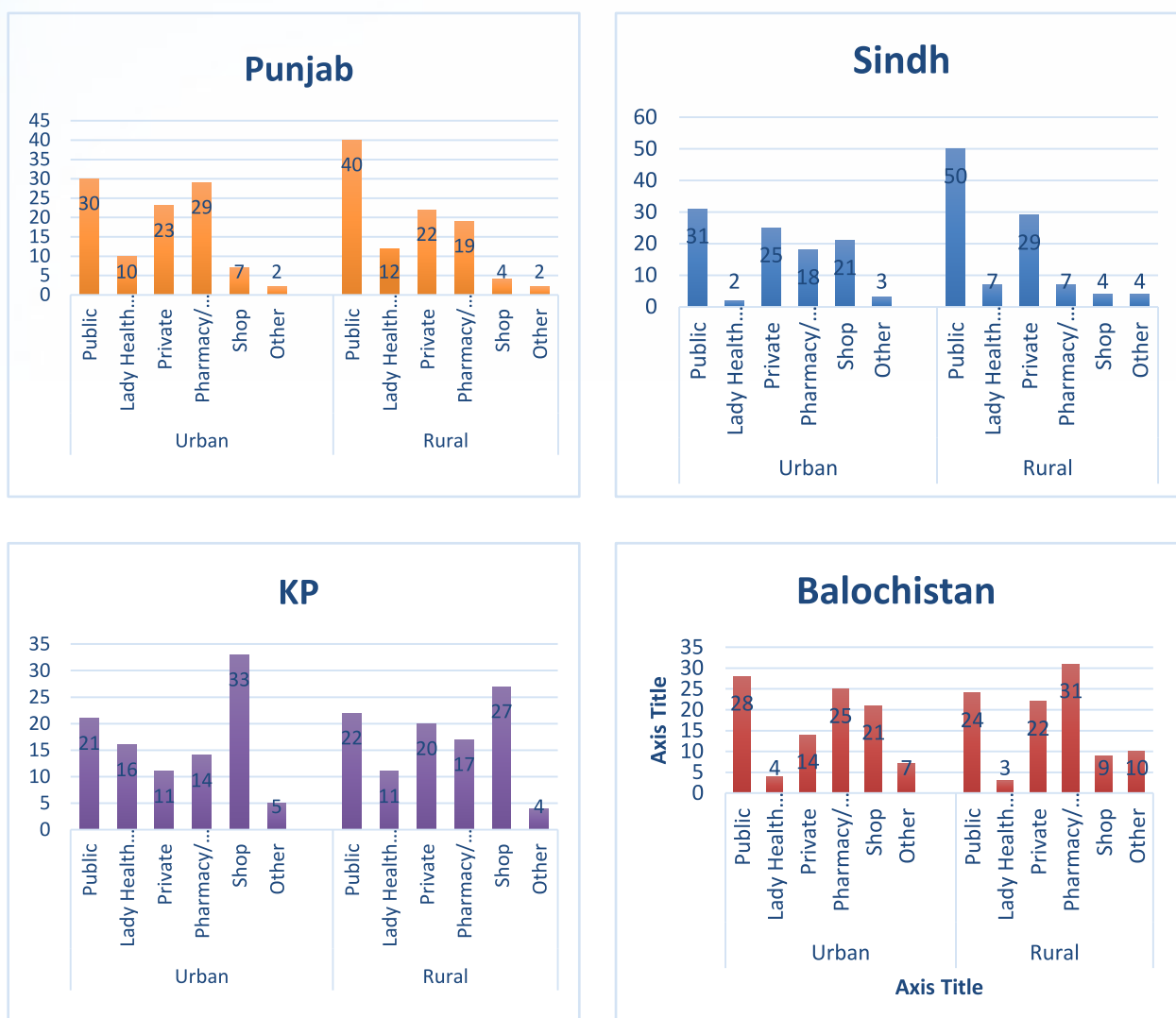
¹⁵ Respondents with no children were asked "If you could choose exactly the number of children to have in your whole life, how many would that be?" Respondents who had children were asked "If you could go back to the time when you did not have any children and could choose exactly the number of children to have in your whole life, how many would that be?"

Knowledge and access to FP services: universal knowledge but access a challenge

Knowledge is not a limiting factor in Pakistan. Knowledge of modern contraceptive methods was nearly universal (over 98% of women and men) and on average respondents in the PDHS 2017/18 listed more than seven methods.

Physical access to FP services is provided by 96,000 lady health workers, 4,000 outlets of the Population Welfare Department and 11,000 facilities of the health department across the country. In addition, private pharmacies and clinics are also providing condoms and hormonal pills and injectable contraceptives¹⁶.

3. Source of supply for current modern method users, by residence and province



¹⁶ Population Council July 2019, "Improving Access to Family Planning Services through the Private Sector in Pakistan – A Stakeholder Analysis" commissioned by the United Nations Population Fund (UNFPA) accessed at https://www.familyplanning2020.org/sites/default/files/Private-Sector-Stakeholder-Analysis-July-15_aa3.pdf

A qualitative study, documented that FP is considered as an additional expenditure by male partners. Besides access, another barrier towards use includes affordability, where frequency of use and choice of method was affected by financial limitation¹⁷.

Financial barriers to access can be gleaned from the fact that the current users of modern FP are significantly higher in the highest income quintile compared the poorest quintile. Modern contraceptive use among currently married women increases according to wealth quintile, from 17% for women in the lowest quintile to 30% for those in the highest quintile. A study by Marie Stopes found that vouchers enabled 10 times more women than expected to choose LARCs (Long Acting Reversible Contraceptives) and PMs (Permanent Methods) in those areas of Pakistan (73,639 vs. 6,455 expected)¹⁸.

Readiness and quality of services: concerns about stockouts

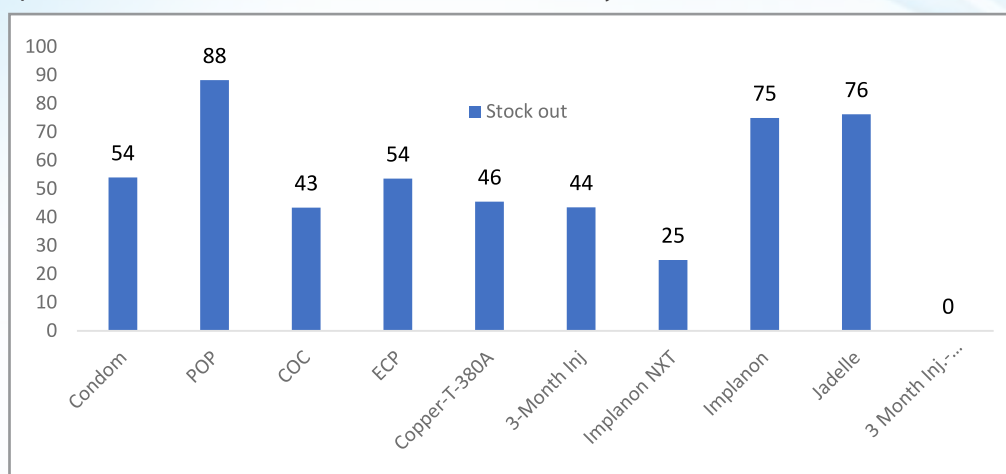
The routine health information systems (DHIS2) can provide information on the availability of family planning commodities at public health facilities. Figure 2 shows the proportion of all public health facilities that had stockouts of specific FP commodities during July 2019 and June 2020. Stockouts were common for nearly all contraceptives, often exceeding half of all reporting health facilities. Injectables had the lowest rate of stockouts but were still as high as 20% of facilities during most months of the year. These high levels of stockouts will influence the choice of contraceptives women have and can lead to disruptions in availability of commodities for users.

Family planning stockouts are attributed to weak procurement capacity of provinces after the 18th amendment and passing of procurement function from federal to provincial levels. At the same time USAID's support for contraceptive supply was curtailed. Contraceptive procurement is fragmented by separate purchasing through each provincial health and population welfare department as well as a separate order for LHWs. Moreover contraceptives are not included in the essential medicines list for buying through MSDs. Low quantities do not allow competitive rates at the international market. Recently the Population unit of the National Ministry has started requesting provinces to share their need for a pooled request for bids to get better pricing estimates.

¹⁷ Azmat SK et al. (2012) Barriers and perceptions regarding different contraceptives and family planning practices amongst men and women of reproductive age in rural Pakistan: a qualitative study. Pak J Public Health 2:1

¹⁸ Luke Boddam-Whetham, Xaher Gul, Eman Al-Kobati, Anna C Gorter. "Vouchers in Fragile States: Reducing Barriers to Long-Acting Reversible Contraception in Yemen and Pakistan" Global Health: Science and Practice Aug 2016, 4 (Supplement 2) S94-S108; DOI: 10.9745/GHSP-D-15-00308

4. Contraceptives stockouts in Public Health facilities (DHIS July 2019-June 2020)



The month-wise DHIS data on stock-outs illustrate long standing stock-outs of all contraceptives in the later half of 2019, even before the spread of COVID-19. Procurement issues and global supply chain disruptions are the likely causes. The situation of injectables is better due to the local production capacity and minimal reliance on imports.

Quality

Different frameworks are used to measure quality of family planning services. One of the frameworks is outlined below for illustrative purposes.

Table 3. Measuring Quality of Family Planning and Client Satisfaction

Facility characteristics	Process	Client satisfaction
Facility with contraceptives inventory	Visual and auditory privacy ensured	Total percentage of clients offered FP counseling and services
Trained providers present	No. of reproductive health and physical examination elements performed	Percentage of clients satisfied with the FP counseling and service
Supervisory visit in last 6 months	Client concerns noted	Percentage of clients satisfied with the FP counseling and service (by age)
System of quality assurance	Confidentiality assured	Percentage of clients satisfied with the FP counseling and service (by education)
Total FP methods offered	Client told about side effects	Percentage of clients satisfied with the FP counseling and service (by income quintile)
Protocols on FP followed	Percentage of clients offered long term FP methods	
FP client record maintained	Waiting time	
Quality stock inventory	Counseling services available both for men and women	

Source: Adapted from Hutchinson et al¹⁹.

¹⁹ Hutchinson, P.L., Do, M. & Agha, S. Measuring client satisfaction and the quality of family planning services: A comparative analysis of public and private health facilities in Tanzania, Kenya and Ghana. *BMC Health Serv Res* **11**, 203 (2011). <https://doi.org/10.1186/1472-6963-11-203>

Pakistan does not have a systematic mechanism to gauge the quality of family planning services. The status of stock outs has already been mentioned above. On the process indicators, limited information is available from PDHS 2017-18, whereby more than three-quarters (78%) of women age 15-49 who are not using a contraceptive method said they did not discuss family planning either with a LHW or at a health facility in the 12 months before the survey (PDHS 2017-18). There has been a substantial decline in women who were visited by LHW and discussed family planning (from 29% in 2012-13 to 19% in 2017-18). By age, women age 30-34 were most likely (24%) and women age 15-19 were least likely (9%) to have been visited by a LHW and discuss family planning. The percentage of women who did not discuss family planning with a LHW or at a health facility is lowest in Sindh (68%) and highest in FATA (95%). Rural and urban women were about equally likely to not discuss family planning at all during their visit.

For informed choice, only 19% of women are informed about all three quality-of-service indicators (side effects, what to do in case of side effects, and other available methods). On average, public sector users were better informed (26%) than private sector users (9%) of all three indicators.

Increasing skilled birth attendance and facility deliveries in Pakistan are a huge opportunity for FP counselling and services. However, only 11% women were given information regarding contraception during postnatal check-up (PDHS 2017-18).

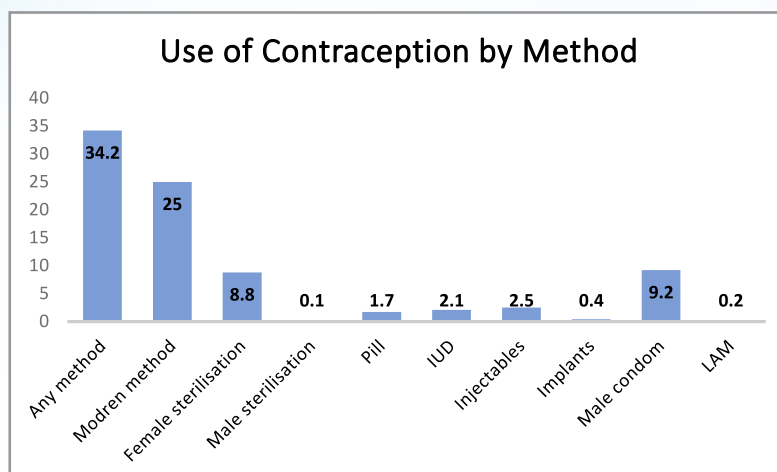
User uptake: low contraceptive prevalence rates

The use of modern contraceptives in Pakistan was low (25.0% of currently married women in PDHS 2017/18), which slightly lower than in 2012/13, and more than 20% lower than in all countries in the South Asia except Afghanistan. The PSLM reported a higher contraceptive prevalence rate among currently married women, 34%, which was slightly up from the previous PSLM 2013-14 (32%). Most of the increase occurred among older women 35-49 years. Both sources agree that there was little increase over time.

Even though there were expected disparities in modern contraceptive use between urban and rural residence, low – high education, poorest – richest and between the regions, the differences were smaller than observed in many other countries. Modern contraceptive use among currently married women 15-49 was only 29% among urban women, 30% among the richest wealth quintile, 30% among women with a higher education, 35% in ICT Islamabad and 27% in Punjab.

Data for this section is primarily derived from two sources i.e., HMIS (Hospital management information system) and PDHS (Pakistan demographic health survey). Data on family planning visits and methods of modern contraceptives are obtain from HMIS and PDHS. Fertility, unmet need and demand for modern contraceptive related data is derived from PDHS.

5. Use of modern contraceptives by method, PDHS 2016/7



Female sterilization and male condoms remained the most common methods among modern contraceptives. Implants and emergency contraception have been added to the method mix more recently. Traditional methods continued to be commonly used, and most of these were withdrawal (8% in 2017/18).

According to the latest FP 2020 self-report survey²⁰, the contraceptive prevalence rate for the year 2019 was estimated to be 36% which presents a major increase since the PDHS 2017/18 but the increase would not be sufficient for the 50% target for 2020. If these data are correct, it would be clear evidence that Pakistan has gained momentum and a sense of realization that population issue needs to be addressed and family planning services should be widely available at all facilities.

Effective coverage: demand for FP satisfied increasing

The demand satisfied with modern methods, which is the FP coverage, is defined as the contraceptive users among the total of women who use modern contraceptives plus those with unmet need (for spacing and limiting). It is an effective coverage indicator because it captures the desired outcome – not being pregnant.

As expected, there was little change in FP coverage between PDHS 2012/13 and 2017/18. The demand satisfied with modern methods increased slightly from 47.0% in PDHS 2012/13 to 48.6% in PDHS 2017/18. All methods increased from 63.8% to 66.4%, showing the continued role of the use of traditional methods. The demand for FP remained low (64% and 65% respectively). The percent of currently married women with unmet need for modern methods declined from 20% to 17%, suggestive of a modest improvement in access. A target of 75% by 2030 is still far.

²⁰ <http://www.familyplanning2020.org/sites/default/files/Pakistan-FP2020-Questionnaire-Self-Report-2019.pdf>

The gaps between urban and rural women were moderate in PDHS 2016/17: 74.2% and 61.0% respectively. Similarly, difference by education (ranging from 59.7% among women with no education to 74.6% among those with higher education) and between the poorest and richest households (47.2% and 76.5% respectively) were observed. The regional disparities ranged from 47.9% in Baluchistan to just over 70% in Punjab and ICT Islamabad.

Increasing demand for FP needs to involve men. Waqas Hameed et al. suggest encouraging family-planning programmes to engage men within the scope of their interventions, as contraceptive use rests more on couples' decisions than on women's decision only. Moreover, efforts need to be made to educate both partners equally about contraceptive methods that have higher effectiveness²¹.

High Abortion rate

Pakistan currently has one of the highest abortion rates in the world. Abortion is becoming the means to exercise birth control, and most of these abortions are being conducted in unsafe environments. In 2002, Pakistani women experienced about 2.4 million unintended pregnancies; nearly 900,000 of these pregnancies were terminated by induced abortion²². 48% of pregnancies in Pakistan are unintended, of which 54% are terminated, mostly in an unsafe way. Around a third of all women who undergo abortions suffer complications, ranging from heavy bleeding to a perforated uterus and deadly infections. A national survey of public-sector facilities estimated that about 200,000 women were hospitalized for abortion complications in 2002²³. In addition, many other women suffered complications but never reached hospitals. Unsafe abortion is also the cause of a substantial proportion of maternal deaths occurring in hospitals. A 1999–2001 university hospital study found that 11% of all maternal deaths that occurred in the hospital during this period were caused by complications resulting from unsafe abortion²⁴. In an earlier hospital study (1992–1994), unsafe abortion had caused 15% of maternal deaths²⁵. While Pakistan's fertility rate is high compared with more developed countries, it is lower than would be predicted by the country's low level of contraceptive use: On average, women have 4.1 live births, and, in metropolitan areas, the rate is just 3.0 live births. It is difficult to explain such fertility rates solely in terms of contraceptive use. Widespread induced abortion is likely to an important contributing factor²⁶. The fact that women in the two less developed provinces—Balochistan and North West Frontier Province—have lower contraceptive use rates and higher abortion rates than women in the more developed provinces, supports this interpretation.

²¹ W Hameed, SK Azmat, M Ali, MI Sheikh, G Abbas... (2014) Women's empowerment and contraceptive use: the role of independent versus couples' decision-making, from a lower middle income country perspective - PloS one, 2014 - journals.plos.org

²² Population Council, *Unwanted Pregnancy and Post-Abortion Complications in Pakistan: Findings From a National Study*, Islamabad, Pakistan: Population Council, 2004.

²³ Sathar ZA, Singh S and Fikree FF, Estimating the incidence of abortion in Pakistan, *Studies in Family Planning*, 2007, 38(1):11–12.

²⁴ Mahmud G and Mushtaq Z, *The Incidence and Outcome of Induced Abortions at One of the Hospitals of Islamabad*, Islamabad, Pakistan: Population Association of Pakistan, 2001.

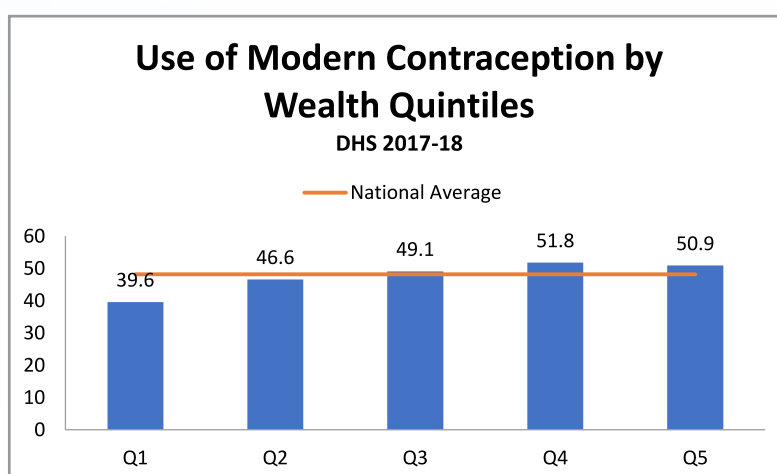
²⁵ Tayyab S and Samad N, Illegally induced abortions: a study of 37 cases, *Journal of the College of Physicians and Surgeons Pakistan*, 1996, 6(2):104–106.

²⁶ <https://www.guttmacher.org/report/abortion-pakistan#>

Who is left behind?

The demand for FP satisfied with modern methods stands at 48.2% nationally, with rural 47.1% and urban 49.9%. It varies by women's education (44.5 for women with no education, 54.7% for those with primary education and 49.8% for those with secondary education and above). The disparity between different income quintiles shows that the poorest women are more disadvantaged than other high income groups.

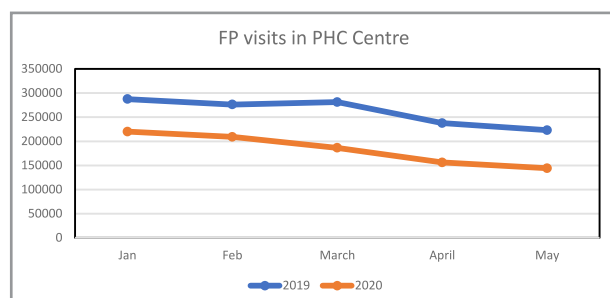
6. Use of modern contraception by wealth quintiles



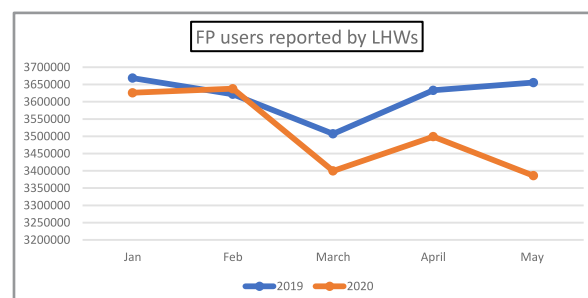
COVID 19 impact

The DHIS data for total FP users at community and PHC services show a negative impact of COVID-19.

7. FP visit in PHC Centre



8. FP users reported by LHWs



Disruption in family planning services could be minimized through a) preparedness actions, b) crisis response, and c) coordinated transition back to routine services.

There is every possibility that due to partial lock downs, facilities may face further stock out of commodities and health workers being overwhelmed with work regarding the other duties assigned for COVID 19 response. These can result in unwanted pregnancies, induced abortions and potential morbidity of unsafe abortions.

Increasing domestic violence is another sequel of the COVID-19 related lock down and economic hardship. Unemployment and inability to feed the children has increased friction between spouses and violence at home. A third of the women reported violence under lockdown. Most women who experienced domestic violence reported humiliation and insults (87%), but in a few cases, there was also physical violence, and the perpetrators were almost always husbands²⁷.

²⁷ Population Council and UNFPA 2019. "Reproductive Healthcare in the Time of COVID-19: Perspectives of Poor Women and Service Providers from Rahim Yar Khan, Punjab"

CHAPTER 2. MATERNAL HEALTH

Table 4: Main points

	Progress	Challenges
Policy context	In 2018, the Government of Pakistan has made a commitment to increase ANC4 and SBA coverage to >90% and reduce MMR by less than 70 by year 2030; Recent initiatives by provincial government to increase access, quality, coverage to maternal health services The cost of maternity care is covered under the “<i>sehat sahulat</i>” programme for impoverished families.	Provincial MNCH programme units have continued through supplementary funding by subsequent PC-1s which have expired or near expiration
Maternal mortality	The 2019 maternal mortality survey shows a decline in MMR to 186 per 100,000 live births, from 276 in the PDHS 2006/07: an average annual rate of reduction of 3.1% per year between 2005 and 2018	To reach the 2030 target of 70 per 100,000 live births an average annual rate of decline of 8.3% per year is required, more than a doubling of the rate of decrease in the last decade
Coverage: ANC, delivery, PNC	ANC coverage increased to 84% (PLSMS 2018/19), more than half of pregnant women made their first visit in the first trimester and half completed at least four visits (PDHS 2017/18).	Major deficits in the quality of care as shown in a 2020 facility survey in ICT and provincial studies and gaps in health workforce Coverage inequalities are still large Postnatal care coverage stagnated at 62%, with rural women and home deliveries left behind
	Institutional delivery coverage increased from 48% to 66% (PDHS 2017/18) and 70% (PSLM 2018/19). About 60% occurred in private facilities according to PSLM, and the share of the public sector is growing	The proportion of deliveries taking place at a health facility was more than twice as high in ICT Islamabad as in Balochistan (84% versus 35%).
	C-section rate increased to 18% in rural Pakistan and to 11% among births to women with no education, suggesting that most women who need a C-section are receiving the intervention.	Considerable overuse of C-section: 46% of births in the richest quintile, 49% of births to women with higher education and 38% of births in private facilities are by C-section, much higher than a WHO estimated threshold of 15%.
Quality of care	There is an increasing realization about sub-optimal quality of care during ANC, care around the time of birth and PNC. A quality of care framework has been developed for RMNCAH.	Quality of care is a concern and respectful maternity care needs more emphasis

Covid-19	Advocacy efforts and guidance on continuation of RMNCAH services is available	Health facility data show a 34.6% decline in MNH services (ANC, deliveries, C-section and PNC) at public facilities during April and a 38.7% decline during May as compared to 2019.
Way forward	Data gaps from private sector services and on causes and drivers of maternal deaths need to be addressed. Areas of improvement include strengthening CRVS, maternal death notification and MPDSR. Priority interventions to improve maternal health include addressing geographical and financial barriers to access; investing in programmatic leadership by creating MNCAH directorates; increasing number of PC facilities providing 24/7 services; strengthening comprehensive emergency obstetric care at DHQ and THQ levels; strengthening birth, death and pregnancy registration; using maternal death reviews and other quality improvement approaches to tackle patient, provider and health system issues by corrective action	

Introduction

This chapter briefly reviews the main policies and targets for maternal health and describes the salient program implementation efforts and larger epidemiological context of maternal mortality trends in Pakistan.

The assessment of progress uses the cascade of effective coverage. Several steps in the cascade have a good level of available data from the national (especially PDHS 2017/18) and provincial surveys (MICS), public sector health facility data, health facility assessments and research. It is possible to identify areas of progress and major challenges. This chapter assesses:

- level and trends of the antenatal, natal and postnatal care
- access to services, readiness and quality of services
- extent to which services are received and used (ANC 4+ and SBA)

Main policies

By 2030, the SDG target (3.1.1) is that the country reduces its maternal mortality ratio to 70 per 100,000 live births or less. In 2018, the Government of Pakistan has made a commitment to increase ANC4 and SBA coverage to >90% and reduce MMR by less than 70 by year 2030.

Program implementation efforts

Pakistan started investing in a network of service delivery infrastructure of Basic Health Units, Rural Health Centers and Tehsil and District level hospitals since the 1960s. Doctors, Nurses, Lady Health Visitors, Family Welfare Workers and Community midwives are qualified as skilled birth attendants. A breakthrough in community based maternal and child health services was observed with the launch of the National Programme for Family planning and Primary Health Care in 1994 which recruited about 90,000 Lady Health Workers. However, the recent programme evaluations have shown a decreasing

performance of LHWs due to resource gaps in financing and technical supervision.

The National MNCH Program²⁸ is being continued with the planned outputs to upgrade hospitals and health facilities in order to provide comprehensive, basic and essential EmONC facilities, train community midwives and improve knowledge and skills of health care providers. The delays and breaches in financial commitment by Government and development partners; loss of operational leadership due to devolution of health sector to provinces under the 18th amendment in 2011; and ambiguity about the deployment of community midwives were some of the challenges encountered²⁹. The program implementation period has shown significant improvement in PDHS (2017-18) for improving antenatal care, use of skilled attendant at birth and access to life saving emergency obstetric care (increase in proportion of caesarean deliveries) and reduction in MMR is noted (MMS 2019).

The Government of Pakistan continues to strengthen maternal and child health care. With devolved healthcare, provincial departments of health have actively led the implementation efforts. The special initiatives launched by the Government of Punjab since July 2019 to December 2019 included scaling up of the Sehat Insaf Card Programme; doubling the size of the LHW programme; increasing number of 24/7 Basic Health Units (BHUs) and Rural Health Centres (RHCs) equipped with a basic package of services, staff and ambulance service; establishing training institutes for nurses and paramedical staff; upgrading secondary care facilities; and building state of the art hospitals in major urban centers of the province. The Government of Khyber Pakhtunkhwa has initiated up-gradation of several BHUs to RHC level in the province from July 2019 to December 2019. Sehat Sahulat Programme has been extended to 69 % of the population of the province. The Government of Balochistan is committed to enhancing health infrastructure/facilities across the province. Construction of one new BHU/RHC in each Tehsil of Balochistan and strengthening of 16 potential DHQs (50 Bedded Hospital) is planned³⁰. Similarly Sindh, AJK, GB and ICT regions have supported MNCH through multiple initiatives.

Provincial MNCH programme units have continued through supplementary funding by subsequent - PC-1s which have expired, or are near expiration. Re-introduction of MNCH directorates in the provincial departments of health is needed to continue the leadership and programmatic support for MNCH programming for achieving SDGs on the reduction of maternal, neonatal and under-five mortality.

General epidemiological trends: decline in maternal mortality but SDG target remain far off

The 2019 Maternal Mortality Survey suggests a decline in maternal mortality to 186 per 100,000 live births, from 276 in the PDHS 2006/07. This corresponds with an average annual rate of reduction of 3.1%

²⁸ <http://dynasoft.org/mnch/overview.php> accessed on 17/06/2020

²⁹ Sarfraz, M., & Hamid, S. (2016). Exploring Managers' Perspectives on MNCH Program in Pakistan: A Qualitative Study. PloS one, 11(1), e0146665. <https://doi.org/10.1371/journal.pone.0146665>

³⁰ Pakistan Economic Survey Report 2019-2020

per year between 2005 and 2018³¹. To reach the 2030 target of 70 per 100,000 live births an average annual rate of decline of 8.3% per year is required, which is more than a doubling of the rate of decrease in the last decade (Table 5).

The pregnancy-related mortality decline is much smaller, as the proportion of deaths due to other causes besides indirect or direct maternal deaths increased considerably. In PDHS 2006-07 16% of 193 pregnancy-related deaths were not maternal deaths. In PMMS 2019, 33% of 179 deaths were not maternal deaths.

The decline was most pronounced in Baluchistan which registered very high maternal mortality levels in PDHS 2006-07. Baluchistan still registered the highest maternal mortality. All four regions need to achieve more than double the rate of decline to reach the 2030 target of 70 (Table 6).

Further disaggregation of maternal mortality data is limited by the relatively small number of deaths, which leads to very large sampling errors. Among adolescent girls 15-19 years maternal mortality was about the same as for all ages (194 per 100,000 live births), and 10% of all maternal deaths took place in this age group. Maternal mortality risks were lowest for women in their twenties and highest for women 35-39 years (481 per 100,000 live births) accounting for one-fourth of all maternal deaths. Women in their forties also had elevated maternal mortality risks but contributed to just 4% of all maternal deaths, as fertility is much lower at this age.

Unfortunately, the causes of maternal deaths were not yet available from the PMMS 2019 Key Indicator Report. PDHS 2006-07 had attributed the highest number of maternal deaths to post-partum haemorrhage, sepsis and eclampsia. Professionals opine that the trends continue with exception of an increasing number of women with pre-existing medical conditions such as hypertension and diabetes which may be adding to more deaths in the category of indirect causes.

Table 5. Maternal mortality

Survey	Mid point of the 3 year period preceding the survey	Pregnancy-related mortality per 100,000 live births	Maternal mortality per 100,000 live births (95% CI)
PDHS 2006-07	2005.5	297	276
PMMS 2019	2018.2	251	186 (138-234)
Average annual rate of change		-1.3	-3.1
SDG	2030.0		70
Average annual rate of change			-8.3

All the provinces and regions have shown a significant progress in reducing MMR since 2006-07.

³¹ 2005 and 2018 are the midpoints of the 3-year period of the retrospective maternal mortality estimates.

Table 6. Region wise MMR and rate of reduction

Province/Region	PDHS 2006-07	PMMS 2019	Average annual rate of change	Average annual rate of change required to reach SDG target for MMR (70)
Punjab	227	157	-2.8%	-6.8%
Sindh	314	224	- 2.5%	-9.9%
KPK	275	165	-3.8%	-7.3%
Baluchistan	785	298	-7.3%	-12.3%
GB		157		
AJK		104		

It is important to note that MMR is higher among women in the age group 15-19 than among 20-24 age bracket. The mortality starts increasing again at older ages: for age 35-39 and is highest among those in 40-44 age group.

Assessment of progress

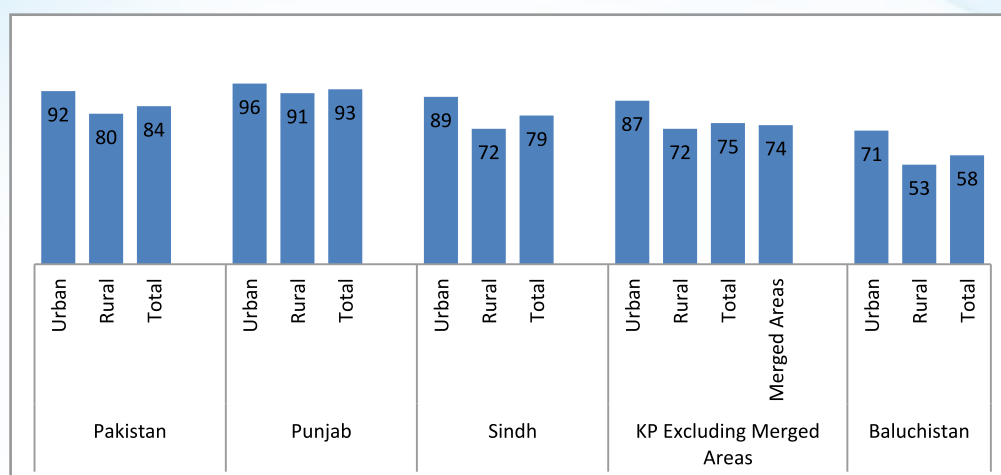
Access to services

Female mobility, distance to health care facilities and purchasing power are some of the factors limiting access. About two-thirds of women reported at least one problem in accessing health care for themselves (67%). About three-fifths of women reported not wanting to go alone (58%), for two-fifths distance to a health facility was a problem (42%), about one-third of women recounted problems getting money for treatment (30%), and one-fifth mentioned that getting permission for accessing health care was a big problem (21%). (PDHS 2017-18)

Antenatal Care Seeking: increasing coverage but gaps in continuity and quality

Antenatal coverage for at least one visit increased from 73% to 86% between 2012-13 and 2017-18 (both estimates for the five-year period preceding the surveys), indicating major progress towards near universal coverage of the first visit of ANC. More than half of pregnant women made their first visit in the first trimester and half completed at least four visits. The Pakistan Social and Living Measurement Survey (PSLM) recorded a similar increase from 72% in 2013/14 to 84% in 2018/19. Rural ANC1 coverage was 80% in the 2018/19 PSLM. Balochistan remained the province with the lowest coverage (58%).

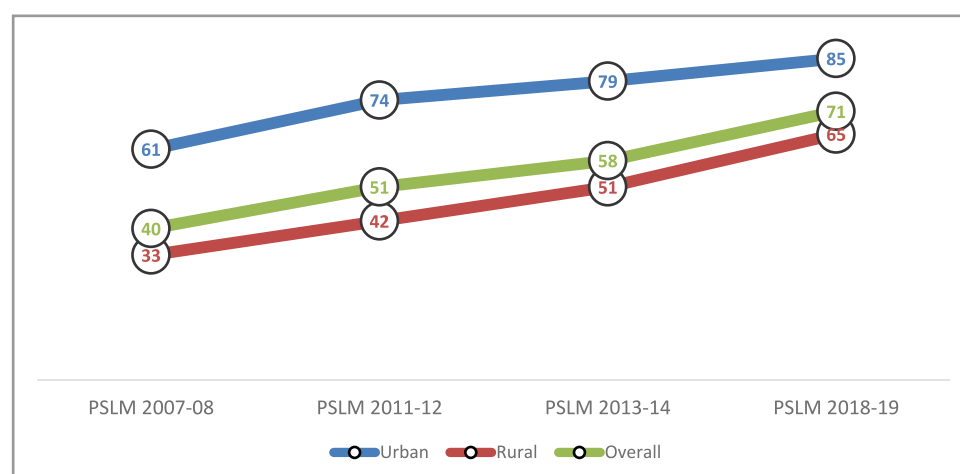
9. Percentage of women age 15-49 who had a live birth in 5 years before the survey (PSLM 2019)



Younger women, ages 15-35, were more likely (85% to 88%) to use ANC services from skilled providers than their older counterparts, ages 35-49 (78%). Similarly, urban women were more likely than rural women to receive ANC from a skilled provider (94% and 82%, respectively). The DHIS2 data comparing total ANC consultations in section on COVID-19 impact, shows that the January 2020 had a higher number than January 2019 but started dipping after COVID-19 related fears and containment measures reduced facility utilization.

Trends in births by skilled birth attendants and place of birth

10. Percentage of live births in the 3 years before the survey –PSLM 2019



The PLSM 2018/19 showed that skilled birth attendance has leap frogged from 51% in 2001 to 71% in 2019. More comprehensive efforts are required to cover the remaining 29%, focusing on bridging the

rural-urban gap. The reliability on government hospitals for childbirth has increased to 29% in 2018-19 as compared to 19% in 2013-14.

PDHS 2017-18 had also shown that coverage of deliveries in health institutions increased to 66% from 48% five years earlier. First-order births are more likely (79%) to occur in a health facility than sixth and higher order births (50%). Eighty-six percent of most recent births to mothers with four or more ANC visits were delivered at a health facility, as compared to only 26% of births to women with no ANC visits. The proportion of deliveries taking place at a health facility was more than twice as high in ICT Islamabad as in Balochistan (84% versus 35%).

About two thirds of all facility births are happening in the private sector though the share of private sector contribution has declined from 70% in 2012-13 to 66% in 2017-18, largely due to increasing number of 24/7 BHUs in Punjab and Sindh. The increase in institutional deliveries is also coupled with a drastic rise in caesarean deliveries from 14 % to 22% in the same period. The C-section delivery rate is higher for births in private facilities (38%) than in public facilities (25%). C-section deliveries are almost twice as prevalent in urban areas (32%) compared with rural areas (18%). Forty-nine percent of births to women with higher education are delivered by C-section, compared with only 11% of births to women with no education. Births to women in the highest wealth quintile are also more likely to be delivered by C-section (46%) than births to women in the lowest quintile (8%). C-section deliveries account for only 3% of births in FATA, as compared with 29% each in ICT Islamabad and Punjab. (PDHS 2017-18)

Postnatal care: persistent gaps

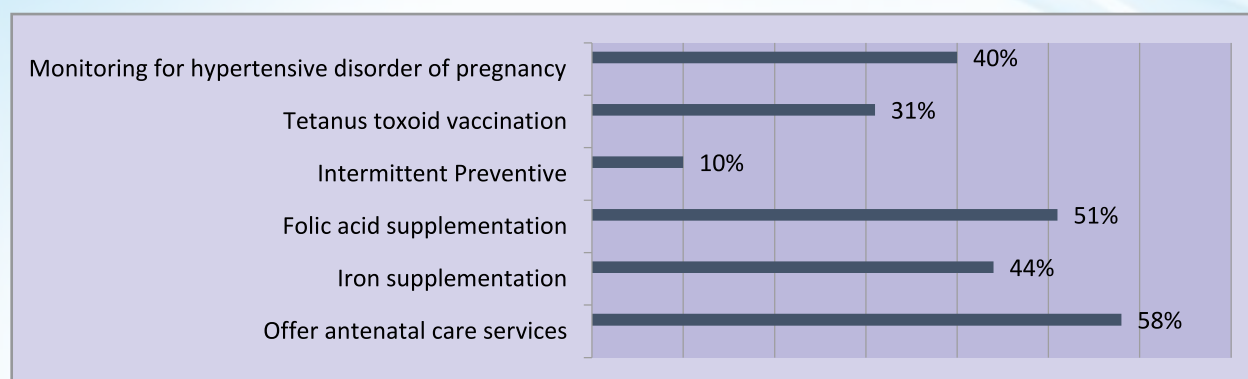
Trends in postnatal care show stagnation in coverage rates and rural-urban disparity. 62% of women age 15-49 who gave birth in the 2 years preceding the survey reported having a postnatal check in the first 2 days after the birth, with most check-ups occurring within 4 hours of delivery (53%). However, 36% of women did not receive any postnatal check during the first 2 days after delivery. The proportion of women with a postnatal check-up within 2 days after delivery has remained largely unchanged between 2012-13 (61%) and 2017-18 (62%). (PDHS 2017-18)

Seventy-nine percent of women who delivered in a health facility received a postnatal check within 2 days after the delivery as compared with only 19% of women who delivered elsewhere. Seventy-six percent of urban women received a postnatal check-up within 2 days after delivery, as compared with 55% of rural women (PDHS 2017-18).

Service availability and readiness: higher gaps at RHC and BHU level

Inadequacies in availability of services may occur due to lack of human resources and/or lack of key commodities and diagnostics. For example, in a 2020 SARA survey in ICT, not all public sector facilities were providing ANC services and gaps were seen in delivery of different interventions.

11. Service availability and readiness



Only one percent of the facilities in ICT had guidelines on ANC and 8% had checklists or job aids; BP apparatus was present in 93% of facilities and adult weighing scale in 85% while only 2% had a haemoglobinometer, 12% urine dipsticks and 30% had iron folic acid tablets. 75% teaching hospitals provided Basic EmOC signal functions as opposed to only 40-60% of RHCs.

DHIS reports (Jan-Dec 2019) show significant vacant positions of Gynecologists, Women Medical Officers (WMOs) and Lady Health Visitors (LHVs).

Table 7. Number of Vacant Positions reported in DHIS 2 (Jan- Dec 2019)

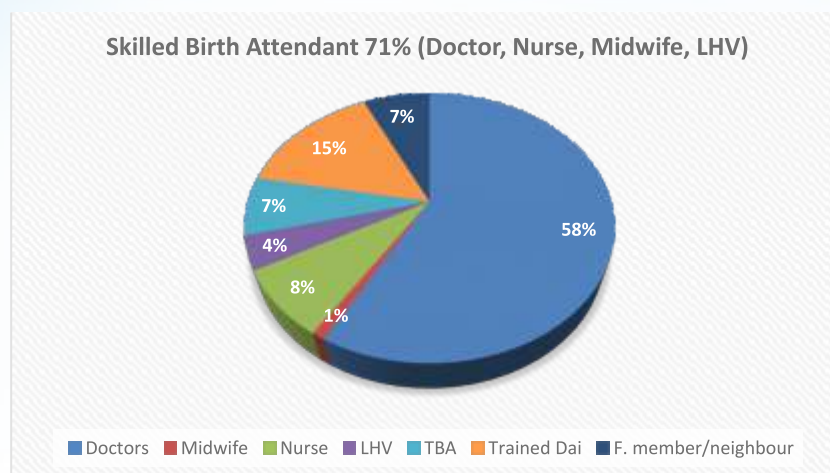
Province/region	WMOs	LHVs
Punjab	96	649
Sindh	3764	357
KP	NA	NA
Baluchistan	456	715
AJK	23	28
GB	1	7

The vacancies exist at PHC level facilities. For example, in Punjab 19 WMO positions were vacant at BHU level and 77 at RHC level. Similarly 556 LHV positions were vacant at BHUs and 93 at RHCs.

Skilled birth attendance: preference for doctors and issues of availability and quality of care

Skilled birth attendance has improved significantly from 52% in 2012-13 to 69% in 2017-18- an increase of 17 points in 5 years' time period (PDHS 2017-18). Recent PSLM 2019 shows skilled birth attendance at 71% with rural coverage at 65% and urban coverage at 85%.

12. Birth Attendant by cadre (PSLM 2019)



Effective quality coverage: issues of prioritization and measurement

Improving quality of care around the time of child birth can save the highest number of maternal and perinatal live. A systematic programmatic intervention and institutionalizing measurement of quality care is very much needed in Pakistan. For example in Punjab, ANC1 coverage was 87%, ANC4 53% but standard ANC 8 was only 15.7%. Among those who had any ANC only 52.6% reported that blood pressure was measured and urine and blood samples were given at least once. (MICS Punjab 2018).

The quality of Antenatal Care needs improvement to ensure all pregnant women get their blood pressure, urine and haemoglobin checked during these visits. Counselling on early initiation of breastfeeding were received by (52%) and on exclusive breastfeeding (54%), while 7 in 10 reported that they had counselling on maintaining a balanced diet during pregnancy. From 2012-13 to 2017-18, the proportion of mothers whose birth was protected against neonatal tetanus increased by 5 percentage points from 64% to 69%.

A recent study from Sindh³² shows that monitoring vital signs during the initial obstetric examination was conducted for less than one-in-ten women. Infection prevention practices were only observed for one-in-four women. Modesty was preserved for less than half of women. In spite of an absence of monitoring during the first and second stages of labour, providers augmented labour with oxytocin in two-thirds of births. To prevent post-partum haemorrhage, oxytocin was administered within a minute of birth in 51% of cases. A support-person was present during labour and birth for 90% of women. Dais or assistants

³² Agha S, Fitzgerald L, Fareed A, Rajbhandari P, Rahim S, et al. (2019) Quality of labor and birth care in Sindh Province, Pakistan: Findings from direct observations at health facilities. PLOS ONE 14(10): e0223701. <https://doi.org/10.1371/journal.pone.0223701>

without formal training provided labour, birth, and newborn care for 40% of deliveries during night shifts at referral facilities.

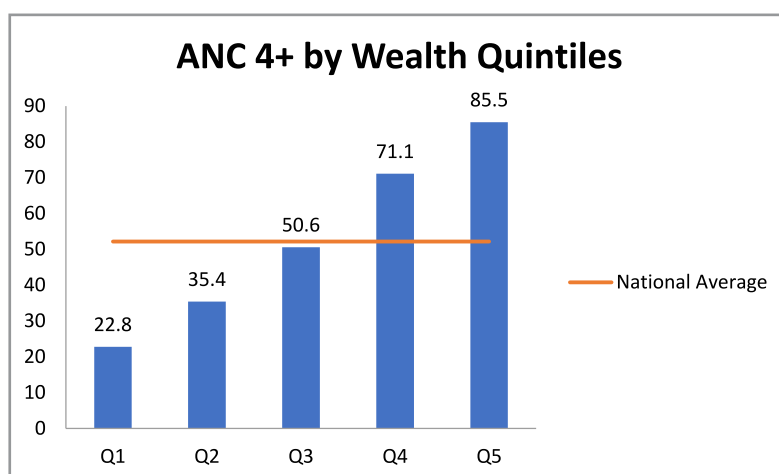
Respectful and dignified healthcare is a fundamental right for every woman. However, many women seeking childbirth services are mistreated by their birth attendants. Approximately 97% of women reported experiencing at least one abusive behaviour. Experiences of mistreatment included non-consented care (81%); right to information (72%); non-confidential care (69%); verbal abuse (35%); abandonment of care (32%); discriminatory care (15%); and physical abuse (15%)³³.

Non-medically indicated caesarean deliveries are a common issue. While there is no gold standard for population C-section rates, WHO has proposed that if C-sections are above 10-15% in the population, it is likely that non-medically indicated (elective) C-section is common. A range of demand and supply factors can play a role. Women may want a C-section based on information they received from health workers, family and friends, or media. Medical staff may favour a C-section for multiple reasons, including their interpretation of risks for normal delivery versus C-section, financial or other incentives. Efforts are underway with support of WHO to promote the use of Robson's classification for understanding if the indications for any caesarean delivery were justified. No study is available from Pakistan.

Who is left behind?

The national average for ANC4+ is 52.2% with rural women having 41% and urban 71% coverage. 78.6% of women with secondary education and above report having 4 ANC visits or more while the ones with primary education 54.3% and no education at 31.8%.

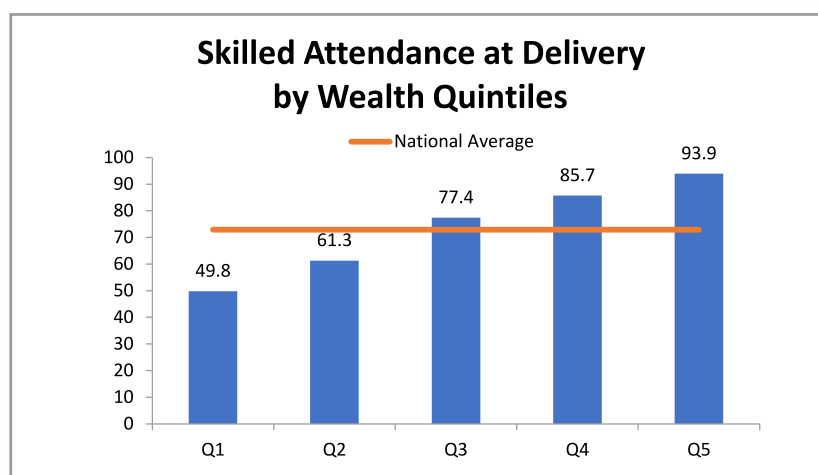
13. 4+ ANC visits by wealth quintiles



³³ Hameed W, Avan BI (2018) Women's experiences of mistreatment during childbirth: A comparative view of home- and facility-based births in Pakistan. PLoS ONE 13(3): e0194601. <https://doi.org/10.1371/journal.pone.0194601>

The national average for skilled attendance at birth is 72.9% with rural women having 66.8% and urban 85.6% coverage. 89.6% of women with secondary education and above report having 4 ANC visits or more while the ones with primary education 74.4% and no education at 60.1%.

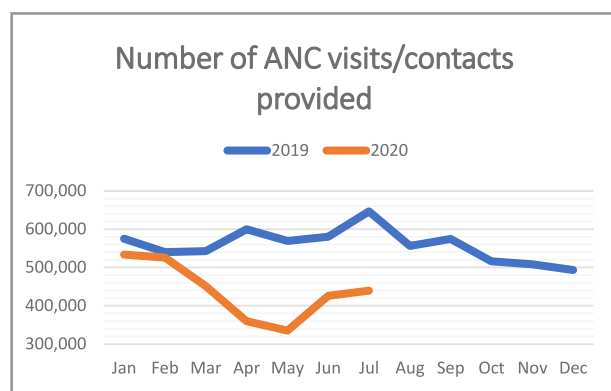
14. Skilled attendance at delivery by wealth quintiles



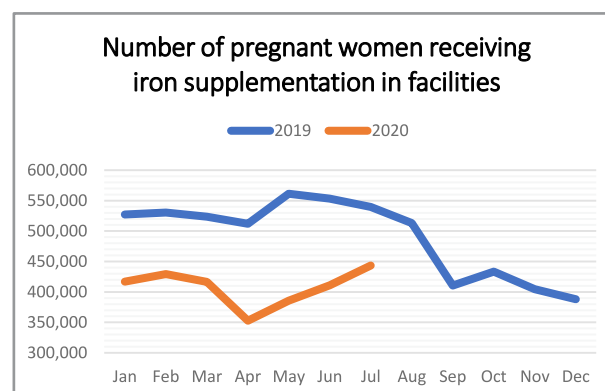
COVID-19 Impact

COVID-19 containment measures like lockdown and closure of some facilities had a negative impact on utilization of services at public facilities. Pakistan is estimated to have 4.7 million births in 2020 according to the UN projections. The average per month is estimated around 400,000 births. Graph no. 16 shows a quarter of all estimated births in public health facilities, highlighting the fact that the private sector is an important player in child birth services.

15. Number of ANC visit/ contacts provided

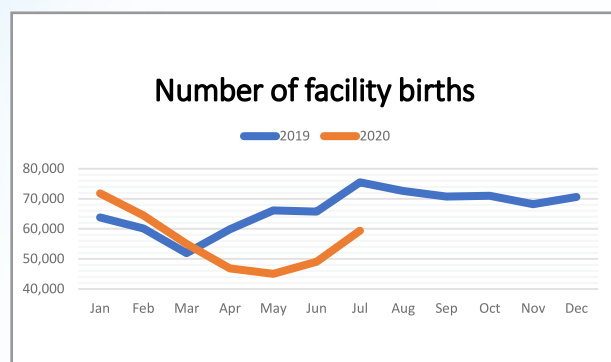


16. Number of pregnant women receiving Iron supplement in facilities

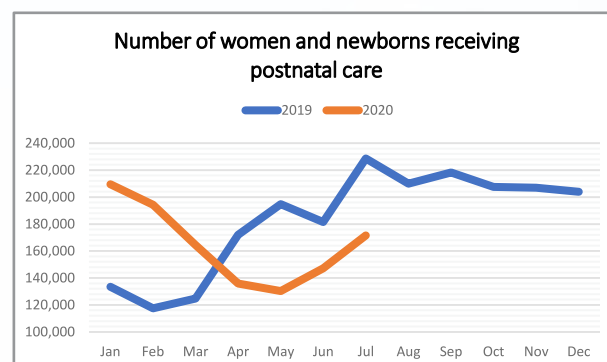


Similar trends were seen for the number of spontaneous deliveries reported from public facilities. Numbers were showing improvement since last year in January 2020 but started falling in the later months due to fears of contracting the infection.

17. Number of facility births

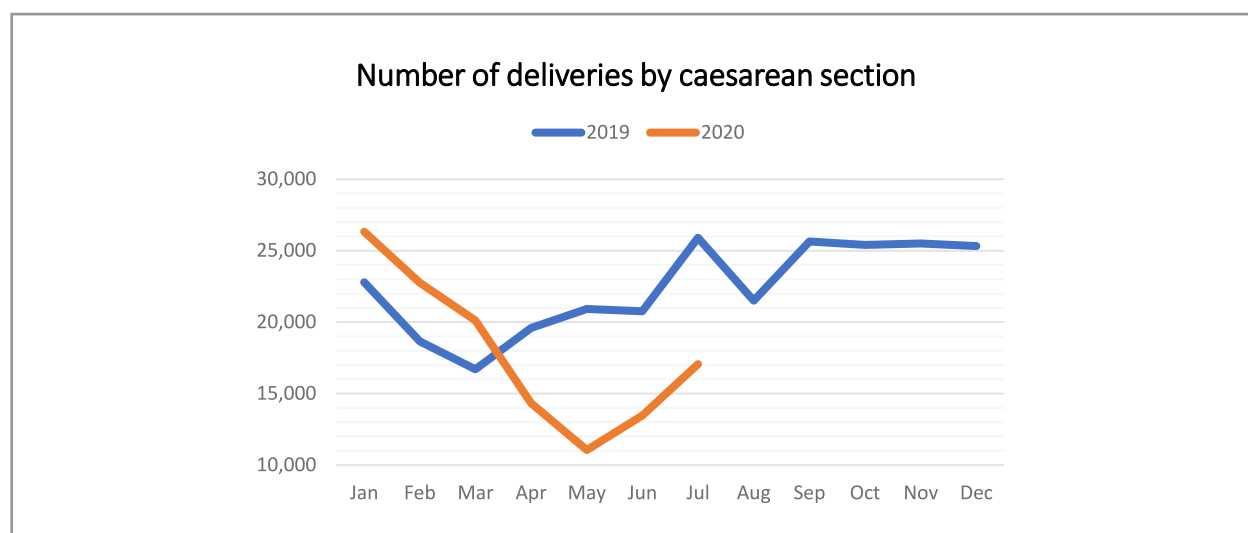


18. Number of women and newborns receiving postnatal care



The numbers of caesarean deliveries showed an upward trend till March as district hospitals remained open. The downward trend in April and May, 2020 was reflective of access issues due to lockdown measures.

19. Number of deliveries by caesarean section



The average decrease in service utilization (for ANC visits, deliveries, C-sections, postnatal care) in public facilities was 6.2% in March, 34.6% in April, 38.7% in May and 21% in June. This analysis helped the Government and partners increase measures for restoring RMNCAH services on a priority basis.

CHAPTER 3. NEWBORN SURVIVAL

Table 8: Main points

	Progress	Challenges
Policy / context	In 2018, the Government of Pakistan has made a commitment to reduce NMR by less than 12 by year 2030 The cost of hospital admissions for small and sick newborns is covered under the “<i>sehat sahulat</i>” programme for impoverished families.	Pakistan has not defined its target for reducing stillbirth rate which is equally high as NMR
Newborn mortality	Newborn mortality decreased to 41 per 1,000 live births in PSLM 2018/19 at an average rate of decline of 2.1% per year during 2010-2019 Research studies show that improving quality of care around the time of birth and focusing on district level small and sick newborn care can reduce neonatal mortality	To reach the neonatal mortality SDG target of less than 12 needs a tripling of the rate of decline to more than 6% per year
Stillbirths	Stillbirth rates are estimated to have declined at 2.0% per year during 2010-2019 to 31 per 1,000 births	Perinatal death reviews are still isolated efforts though preliminary quality of care efforts have been initiated in selected tertiary hospitals.
Coverage	Programmatic and financial inputs are provided for provision of small and sick newborn care and improving quality of care	Data on key interventions such as the use of bag and mask to resuscitate newborns and Kangaroo mother care (KMC) for babies < 2000 grams not available; Still births and neonatal deaths by cause are not reported in DHIS
	PNC visit made for 64% newborns	6 PNC signal functions are examined in a minority of cases – poor quality
Quality of care	National guidelines on KMC have been developed	KMC is used in only a limited number of facilities UNICEF/ AKU survey of hospitals on small and sick newborn care shows major quality of care issues; ICT facility assessment in 2020 suggest poor use of interventions such as bag and mask

Way forward	A national neonatal survival strategy, costed plan and adequate financing is critical to accelerate progress for achieving the SDG target. Human Resource strategy and plan for developing cadres of neonatologists, neonatal nurses and auxiliaries for neonatal intensive care units are urgently required. Priority interventions to improve neonatal survival and still birth reduction include financing of the provincial costed plans, strengthening data systems including MPDSR, addressing HR gaps and improving quality of care around the time of birth and for small and sick newborns. Surveillance of congenital anomalies and efforts to measure and reduce still births are needed. MDSR activities need to add 'P' for MPDSR to understand the determinants of neonatal mortality at each facility and district for planning corrective actions. DHIS reports need more specificity for recording and reporting peri-natal events. Total consultations for under-five children are not disaggregated for newborns. Still births are not captured in DHIS, neonatal deaths are not classified by sex, cause, birth weight or gestational age. Chlorhexidine, ambu bags and masks, injectable antibiotics and oxygen are not included in the category of medicines and commodities reported for stock outs. Coverage of small and sick newborn care including Kangaroo Mother Care is not reported. These caveats need priority action to streamline efforts for newborn survival.
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Introduction

This chapter briefly reviews the main policies and targets for neonatal health and describes the main program implementation efforts and larger epidemiological context of neonatal mortality trends in Pakistan.

The assessment of progress uses the cascade of effective coverage. Several steps in the cascade have a good level of available data from national surveys (especially PDHS 2017/18, PSLM 2018/19 and MICS), public sector health facility data, health facility assessments and research. It is possible to identify areas of progress and major challenges. This chapter assesses:

- level and trends of the neonatal mortality
- access to services, readiness and quality of services
- extent to which services are received and used (PNC) and
- quality of essential newborn care and for the small and sick newborns

Main Policies

By 2030, the SDG target (3.2.2) is that the country reduces its Neonatal Mortality Rate to less than 12 per 1000 live births. In 2018, the Government of Pakistan has made a commitment to reduce NMR by less than 12 by year 2030.

Program Implementation Efforts

Pakistan benefitted from being in the Saving Newborn Lives (SNL-1) with considerable advocacy efforts in early 2000s. Capacity building for essential newborn care, birth asphyxia management and nation-

wide tetanus vaccination campaigns for women of reproductive age was conducted. Later, National MNCH programme scaled up essential newborn care through health education by Lady Health Workers and capacity building of service providers at the facility level. Pakistan was also one of the first countries to develop a national plan following the launch of Every Newborn Action Plan (ENAP) in 2014. A bottleneck analysis exercise highlighted issues of missing /low coverage of interventions like sepsis management and care for small and sick newborn; issues of health financing and information systems. Evidence-based neonatal care interventions were promoted and provincial ENAP plans were developed. However the momentum slowed down due to lack of adequate financing and eighteenth amendment related changes. Capacity building efforts for promoting use of ambu bag and mask for asphyxia management, early essential newborn care and use of chlorhexidine for cord care continued mostly with the support of development partners. Kangaroo Mother Care for managing stable newborns less than 2000 grams was initiated in one of the teaching hospitals in Lahore after an exposure visit to Vietnam in 2014. The practice has been adopted by some hospitals on voluntary basis without additional logistical support. National guidelines on KMC have been developed in 2019 and uniform implementation is being ensured with help of partners.

An assessment on small and sick newborn care was conducted in 2019 in selected hospitals of Pakistan and has revealed stark gaps in care for small and sick newborns. Perinatal death reviews are still isolated efforts though preliminary quality of care efforts have been initiated in selected tertiary hospitals. The cost of maternity care and hospital admissions for small and sick newborns is covered under the “sehat sahat” programme for impoverished families.

Research efforts have included implementation of a set of low cost community-based intervention packages carried out by Lady Health Workers. The study has shown significant improvements in household newborn care practices and reduction in neonatal mortality rates^{34,35}. Another cluster-randomised trial demonstrated that community intervention by the LHW programme led to a 20% reduction in neonatal mortality and was associated with significant improvements in household practices and newborn care practices. However, overall effective coverage by functional LHWs was only 48%, and they were able to perform resuscitation in only 4% of potentially eligible neonates with birth asphyxia. The intervention had no effect on cause-specific neonatal mortality due to asphyxia or suspected serious infections. No effects on care seeking for facility births and stillbirths in the intervention clusters. The implementation of a set of low cost community-based intervention package within the health system settings in Gilgit region was found to be feasible and beneficial. Significant reductions in perinatal and neonatal mortality rates were noted. Current efforts for implementing home-

³⁴ Zulfiqar A Bhutta, Sajid Soofi, Simon Cousens, Shah Mohammad, Zahid A Memon, Imran Ali, Asher Feroze, Farrukh Raza, Amanullah Khan, Steve Wall, Jose Martines, Improvement of perinatal and newborn care in rural Pakistan through community-based strategies: a cluster-randomised effectiveness trial, *The Lancet*, Volume 377, Issue 9763, 2011, Pages 403-412, ISSN 0140-6736, [https://doi.org/10.1016/S0140-6736\(10\)62274-X](https://doi.org/10.1016/S0140-6736(10)62274-X). (<http://www.sciencedirect.com/science/article/pii/S014067361062274X>)

³⁵ Memon, Z.A., Khan, G.N., Soofi, S.B. et al. Impact of a community-based perinatal and newborn preventive care package on perinatal and neonatal mortality in a remote mountainous district in Northern Pakistan. *BMC Pregnancy Childbirth* 15, 106 (2015). <https://doi.org/10.1186/s12884-015-0538-8>

based newborn care in selected districts have also shown promising results (UNICEF).

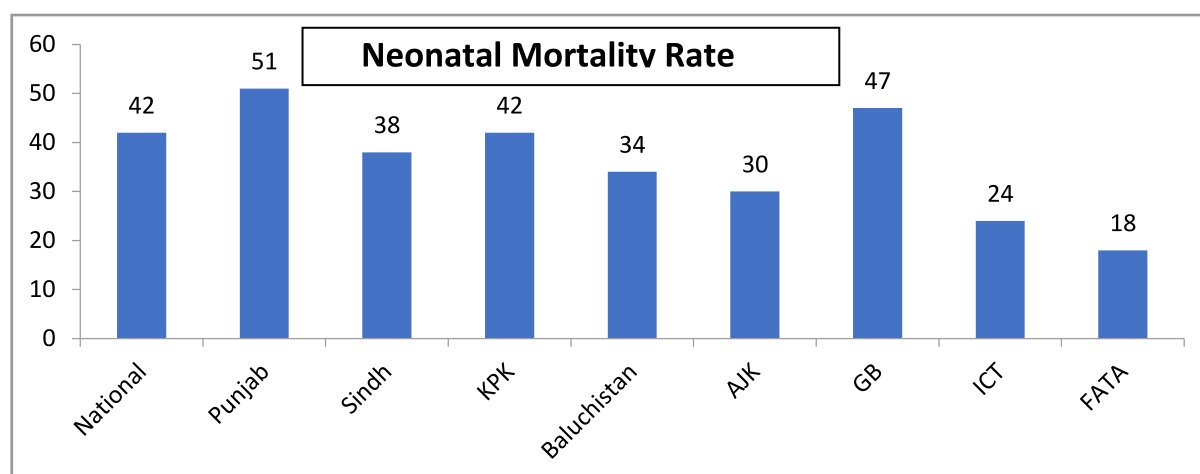
Provincial costed action plans for newborn survival have been developed for Punjab and Baluchistan while newborn care strategies have been developed for Sindh and KPK. The focus on care for small and sick newborns is gaining momentum. The Ministry of National Health Services Regulation and Coordination together with partners has developed guidelines on KMC and included the intervention in the district EPHS package and plans to include it in LHWs curriculum. Sixty two percent of all deaths in Pakistan are attributed to neonatal causes (IHME 2017). Low birth weight, poor quality of care around the time of birth and birth spacing less than two years are the major risk factors for preventable newborn deaths. Neonatal care remains the mainstay for success of RMNCAH plan and UHC benefit package.

General Epidemiological Trends: Slight Recent Improvement in Neonatal Mortality Reduction

Pakistan has witnessed a slow reduction of NMR from 52 per 1000 live births in 2006-07 to 42 per 1000 live births in 2017-18. PSLM 2018/19 report has quoted NMR of 41 per 1000 live births. The UN-IGME estimates used all the survey data on neonatal mortality were for 2019 (41.2 per 1,000 live births) and estimated that neonatal mortality declined at 2.1% per year during 2010-2019, with no acceleration within this period. At the current rate of progress, Pakistan's projected NMR would be 32 per 1,000 live births, far above the SDG target of <12, by the year 2030. The rate of decline will need to triple to at least 6% decline per year to reach the 2030 target.

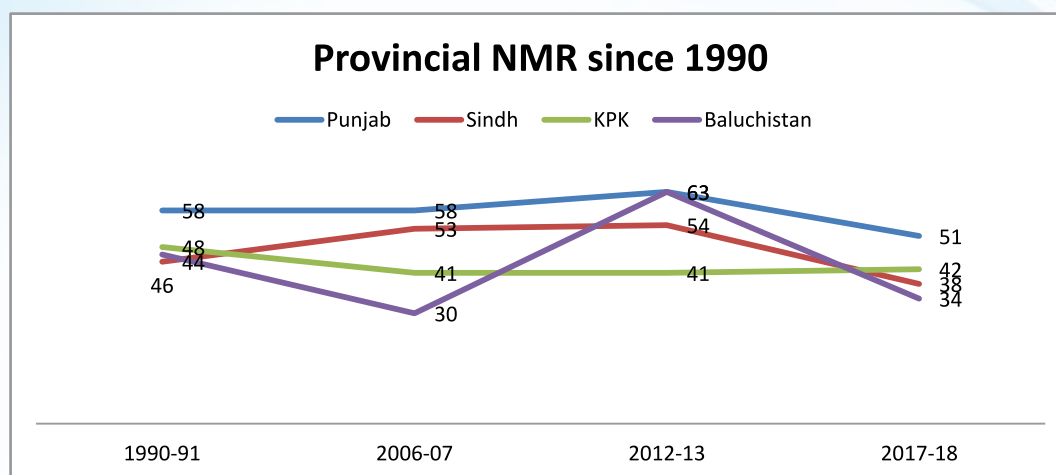
There is a considerable variation in NMR by province:-

20. Provincial variation in NMR (PDHS 2017-18)



All provinces and regions experienced either stagnation or increase in neonatal mortality rates till 2012-13. However, the last PDHS 2017-18 noted a decline in NMR for all the regions. This downward trend was also documented in Punjab MICS 2017-18.

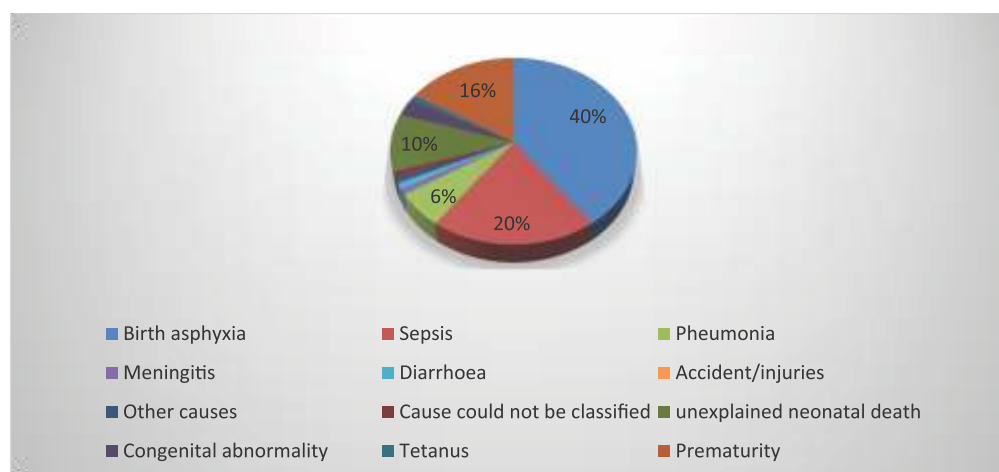
21. Provincial NMR since 1990



Inadequate financing for newborn care and lag in addressing key social determinants such as poverty, female education and empowerment and unregulated population growth has been quoted for compromising progress for neonatal survival in Pakistan³⁶.

Birth asphyxia, sepsis and prematurity were the leading causes for neonatal mortality in PDHS 2006-07 verbal autopsies³⁷.

22. Percent distribution of neonatal deaths by cause of death, Pakistan 2006-07



³⁶ Arjumand Rizvi, Zaid Bhatti, Jai K. Das & Zulfiqar A. Bhutta (2015) Pakistan and the Millennium Development Goals for Maternal and Child Health: progress and the way forward, Paediatrics and International Child Health, 35:4, 287-297, DOI: 10.1080/20469047.2015.1109257

³⁷ NIPS/Pakistan, and Macro International. 2008. Pakistan Demographic and Health Survey 2006-07. Islamabad, Pakistan: NIPS/Pakistan and Macro International

Another study³⁸ concluded that neither delivery in a health facility nor by health professionals was associated with fewer neonatal deaths. Almost all (88%) neonates who died received treatment and 75% died in the hospital. A high neonatal mortality rate, often due to preventable conditions was found in an urban population with good access to professional care suggesting the need to improve health service quality to decrease neonatal mortality. A systematic review^{39,40}, suggested a 29% reduction in NMR with increase of facility deliveries. Comprehensive emergency obstetric care and basic emergency obstetric care resulted in a reduction of intra-partum related neonatal deaths by 85% and 40% respectively⁴¹. A lower reduction in NMR in Pakistan vis-à-vis the increase in proportion of facility deliveries is likely to be due to poor quality of care.

Still Birth Rate

Pakistan's still birth rate has been estimated as 30.6 per 1000 births with 190,483 still births a year⁴². The average annual rate of decline between 2010 and 2019 was 2.0%. (the PDHS 2017/18 had a perinatal mortality of 57 per 1,000 births which included a stillbirth rate of 23.5 per 1,000, but underreporting of stillbirths in common in surveys).

Table 9. Estimates of Still birth rate (UN-IGME)

Still Birth Rate			Percentage decline(percent) 2000-2019	Annual Rate of Reduction (ARR) percent 2000-2019
2000	2010	2019		
39.8	36.5	30.6	23.2	1.4

Number of Still births

Table 10. Number of still births

2000	2010	2019	Proportion of global number of still births
204,880	204,648	190,483	9.7

³⁸ Jehan I, Harris H, Salats, et al. Neonatal mortality, risk factors and causes: a prospective population-based cohort study in urban Pakistan. *Bull World Health Organ* 2009; **87**: 130-138

³⁹ Tura, G., Fantahun, M. & Worku, A. The effect of health facility delivery on neonatal mortality: systematic review and meta-analysis. *BMC Pregnancy Childbirth* **13**, 18 (2013). <https://doi.org/10.1186/1471-2393-13-18>

⁴⁰ Vivek V. Shukla, Wal demar A. Carlo. Review of the evidence for interventions to reduce perinatal mortality in low- and middle-income countries. *International Journal of Pediatrics and Adolescent Medicine* Volume 7, Issue 1, March 2020, Pages 4-10

⁴¹ A.C. Lee, S. Cousens, S.N. Wall, S. Niermeyer, G.L. Darmstadt, W.A. Carlo, et al. Neonatal resuscitation and immediate newborn assessment and stimulation for the prevention of neonatal deaths: a systematic review, meta-analysis and Delphi estimation of mortality effect *BMC Publ Health*, 11 (3) (2011), p. S12

⁴² A Neglected Tragedy: The global burden of still births- Report of the UN Inter-agency Group for Child Mortality Estimation, 2020

Assessment of progress

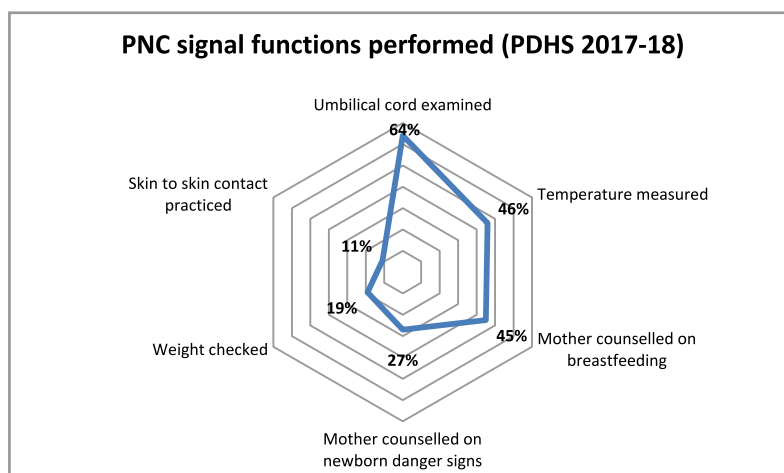
Postnatal Care Seeking: increasing coverage but gaps in quality

PDHS (2017-18) showed that 64% of newborns received a postnatal check within the first 2 days after birth. Two in five newborns (41%) had a postnatal check within the first hour of life. Only 34% of newborns residing in Balochistan received postnatal care during the first 2 days after birth, compared with 77% of newborns in ICT Islamabad.

Fifty-nine percent of newborns received their first postnatal check from a doctor, nurse, midwife, Lady Health Visitor, or community midwife. The proportion of newborns with a postnatal check by these providers was higher among first births (69%), those whose mothers had higher education (85%), and those born to mothers in the highest wealth quintile (86%).

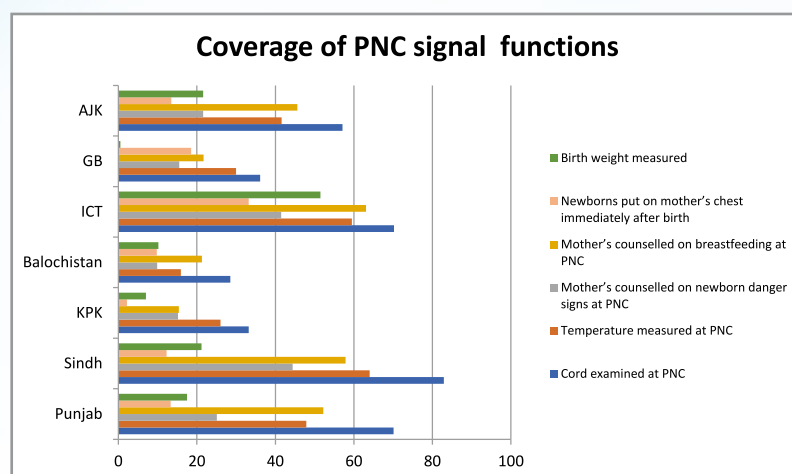
However the quality of postnatal care contacts has shown gaps in content of care and the signal functions performed. Of all the neonates receiving PNC, sixty-four percent of infants born in the 2 years preceding the survey had their umbilical cord examined, and 46% had their temperature measured. Forty-five percent were counselled on breastfeeding, and more than one fourth of mothers received counselling on newborn danger signs (27%) and were observed while breastfeeding (26%). Only 19% of newborns were weighed during the first 2 days after their birth. Fifty eight percent of newborns receiving a PNC contact had at least two signal functions performed.

23. PNC signal functions performed (PDHS 2017-18)



Similar provincial variations for PNC signal functions are also seen whereby all regions show a higher coverage for cord examination but low on skin to skin contact and measuring weight at birth.

24. Coverage of PNC signal functions



While WHO guidelines recommend four PNC visits, there is a lack of understanding and clear programmatic guidance on number of visits, timing and reporting for health providers. PNC checklists were developed for LHWs but no updates are available on their use. Currently, no additional home visits are specified for low birth weight (LBW) and pre-term newborns. Pre-discharge PNC for newborns delivering at health facilities needs improvement with explicit guidelines and job-aids.

Service availability and readiness: gaps in essential newborn care and facilities for small and sick newborn care

Pakistan needs to align its service packages to the recent UNICEF and WHO guidance for setting up essential, specialized and intensive care for newborns. The Universal Health Care district implementation package provides an immediate opportunity to include and scale-up the required neonatal care interventions at appropriate levels of care.

Table: Inpatient care for small and sick newborns: requirements for care at different health system levels⁴³*Table 11: Standards of care and evidence-based interventions*

Level	Type of care provided	Standards of care and evidence based interventions
Primary	Essential newborn care	Immediate newborn care (through drying, skin-to-skin contact of the newborn with the mother, delayed cord clamping, hygienic cord care); neonatal resuscitation (for those who need it); early initiation and support for exclusive breastfeeding; routine care (Vitamin k, eye care and vaccinations, weighing and clinical examinations); prevention of mother-to-child transmission of HIV; assessment, management and referral of bacterial infections, jaundice and diarrhea, feeding problems, birth defects and other problems; pre-discharge advice on mother and baby care and follow-up.
Secondary	Special newborn care	Thermal care; comfort and pain management; kangaroo mother care; assisted feeding for optimal nutrition (cup feeding and nasogastric feeding); safe administration of oxygen; prevention of apnoea; detection and management of neonatal infection; detection and management of hypoglycemia, jaundice, anemia and neonatal encephalopathy; seizure management; safe administration of intravenous fluids; detection and referral management of birth defects. Transition to intensive care: continuous positive airway pressure; exchange transfusion; detection and management of Necrotizing Entero-Colitis (NEC); specialized follow-up of infants at high risk (including preterm).
Tertiary	Intensive newborn care	Advanced feeding support (e.g. parenteral nutrition); mechanical/assisted ventilation, including intubation; screening and treatment for retinopathy of prematurity; surfactant treatment; investigation and management of birth defects; pediatric surgery; genetic services.

Sepsis is a major cause of neonatal death but stark deficiencies exist in managing PSBI (Possible Suspected Bacterial Infections). In Punjab, only 10.9% medical officers and 2.7% other relevant staff at PHC Facilities were trained on Integrated Management of Newborn and Childhood Illness (IMNCI) guidelines. Availability of medicines varied from 35% in primary facilities to 84% in referral facilities⁴⁴.

Facility readiness was also very low in ICT (SARA Survey 2019), whereby 31 facilities varying from tertiary hospitals to BHUs showed the following percentage availability of newborn care services.

Table 12. Facility readiness in ICT (SARA Survey 2019)

Antibiotics for preterm or prolonged PROM to prevent infection	Neonatal resuscitation with bag and mask	Corticosteroids in preterm labour	KMC (Kangaroo mother care) for premature/ very small babies	Injectable antibiotics for neonatal sepsis	Mean availability of newborn signal functions offered
7%	6%	3%	3%	5%	5%
Guidelines for essential childbirth care	Check-lists and/or job-aids for essential childbirth care	Guidelines for essential newborn care	Staff trained in essential childbirth care	Staff trained in newborn resuscitation	Guidelines and trained staff readiness score
1%	1%	n/a	2%	2%	2

⁴³ Survive and thrive: transforming care for every small and sick newborn. Geneva: World Health Organization; 2019. Licence: CC BY-NC-SA 3.0 IGO.

⁴⁴ PSBI baseline Assessment Report 2018, Bureau of Statistics BOS, Govt. of Punjab

75% of tertiary hospitals in ICT were providing counselling on immediate and exclusive breastfeeding compared to 60% RHCs.

The AKU and UNICEF study on Situation Analysis on Stillbirths, Newborn Deaths and Small & Sick Newborn Care 2019, was a first systematic attempt to understand the gaps in coverage and quality of interventions for the small and sick newborns. In a sample of 23 facilities across the country providing in-patient care for newborns (including teaching hospitals), 100% had paediatricians while only 13% had neonatologists, 9% had neonatal surgeons and 13% had neonatal nurses. On a closer examination, the teaching hospitals have consultant neonatologists but there are no positions for this speciality at DHQ and THQ level. Similarly neonatal nursing is an under-developed specialization in Pakistan⁴⁵.

Availability of services for low birth weight and pre-term infants, like Kangaroo Mother Care, is limited to a few tertiary hospitals, presently. National guidelines on KMC have been developed. However, the intervention is still not captured in DHIS and coverage of eligible neonates cannot be determined. Post-discharge follow-up has been rated as one of the biggest challenges of the KMC units. Linkages with local BHUs and follow-up by LHWs needs to be developed. Propositions to scale-up KMC include three pathways: 1) champion-led approach in which a small group led by community person initiates and maintains the KMC intervention; 2) the project-initiated approach focused on project driven and donor funded implementation of KMC and 3) the approach of health systems redesigning incorporated KMC in the existing health system structure. A study had advocated that the combination of all three pathways may lead to a more rapid scale-up of KMC even in resource constrained settings, such as rural Pakistan.

"A newborn's weight at birth is an important marker of maternal and fetal health and nutrition. Low birth weight newborns have a higher risk of dying in the first 28 days of life. Those who survive are more likely to suffer from stunted growth and lower IQ. The consequences of low birth weight continue into adulthood, increasing the risk of adult onset chronic conditions such as obesity and diabetes⁴⁶."

According to the PDHS 2017-18, 22% of all live births had low birth weight. One-third (34%) of live births to mother's under age 20 had a low birth weight compared with one fifth (21%) of births to mothers age 20-34. Babies of high birth order are more likely to be of low birth weight. Twenty-five percent of live births with a birth order of 6 or higher had a low birth weight compared with 21% of those with a birth order of 2-5. Similarly, babies born to mothers who smoke cigarettes were more likely to have a low birth weight than babies born to mothers who did not smoke (28% and 22%, respectively). Babies born in households in the lowest wealth quintile were more likely (33%) to be of low birth weight than those born in households in the highest quintile (19%). Punjab had the highest percentage of babies (24%) with a

⁴⁵ Hasan Nawaz Tahir, Zafar Fatmi (2019) Kangaroo mother care: opportunities and implications for rural Pakistan, J Pak Med Assoc 69 (11):1725-1729

⁴⁶ United Nations Children's Fund (UNICEF), World Health Organization (WHO). UNICEF-WHO Low birth weight estimates: Levels and trends 2000–2015. Geneva: World Health Organization; 2019 Licence: CC BY-NC-SA 3.0 IGO.

reported birth weight of less than 2.5 kilograms. It is closely followed by Sindh at 23%. In Pakistan, 860,000 babies are born too soon (preterm: <37 weeks gestation) each year and 101,600 children under five die due to direct preterm complications⁴⁷.

Effective quality coverage: issues of prioritization and measurement

Increasing facility births have not led to a proportional decline in the neonatal mortality in Pakistan. Poor quality of care at birth and during the postnatal period is evident in maternal counselling, skin to skin contact after birth and measuring weight at birth, as they are particularly low in all provinces and regions.

Table 13. Key maternal and newborn indicators regionswise

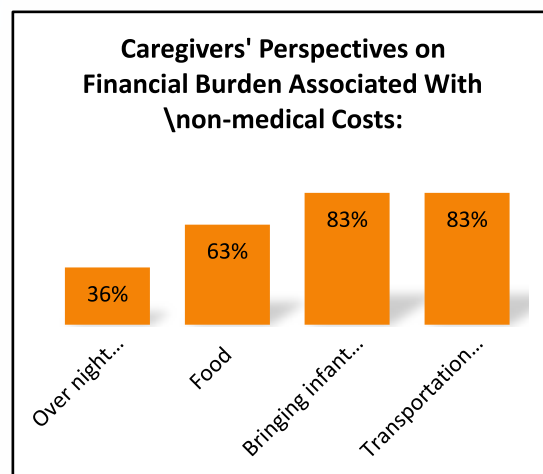
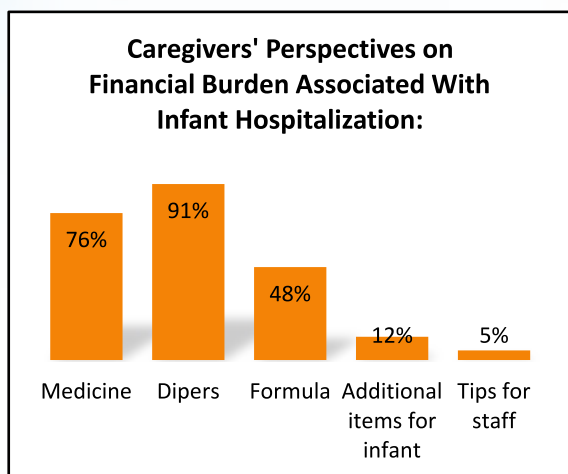
Indicator	National	Punjab	Sindh	KPK	Balochistan	ICT	GB	AJK	FATA
Neonatal Mortality Rate	42	51	38	42	34	24	47	30	18
Perinatal Mortality Rate	57	62	52	53	54	45	43	53	23
PNC Newborn within 48 hrs after birth %	63.9	69	75.1	43.3	34.3	77	38.9	59.9	38.6
PNC(N) by doctor, nurse, LHV or CMW	59.4	65.1	68.9	38.8	29.0	75.0	38.2	53.2	36.7
Cord examined at PNC	64.1	70.1	82.9	33.2	28.5	70.2	36.1	57.1	26.7
Temperature measured at PNC	46.1	47.9	64.0	26.0	15.9	59.5	30.0	41.6	24.9
Mother's counselled on newborn danger signs at PNC	26.9	25.1	44.4	15.2	9.9	41.5	15.5	21.6	6.2
Mother's counselled on breastfeeding at PNC	45.2	52.2	57.9	15.4	21.3	63.1	21.7	45.6	9.7
Newborns put on mother's chest immediately after birth	11.0	13.3	12.3	2.2	9.8	33.2	18.5	13.5	0.6
Birth registration	42	57.8	27.8	18.8	37.6	82.4	27.1	29	2.2
Birth weight measured	16	17.5	21.2	7.0	10.2	51.5	0.5	21.6	22
Low birth weight	22	23.7	22.5	12.8	11.8	18.3	17.4	19.5	-

The UNICEF and AKU study on small and sick newborn care revealed that most of the facilities lacked Kangaroo Mother Care Units and do not routinely undertake maintenance of incubators, radiant warmers and phototherapy lights; only 13 % facilities had a training of staff on counseling of parents; 39% of the facilities were designed as "baby friendly"; 35% facilities monitored their nosocomial infection rates; and approximately 40% of the facilities lacked authority over budget. Majority of the facilities did not conduct inter-disciplinary (87%) and management team meetings (83%). Perinatal and neonatal death reviews was not a common practice across 57% of the surveyed facilities. Patient record reviews showed that none of the facilities record all essential aspects in newborn assessment and admission history and none

⁴⁷ <https://www.healthynewbornnetwork.org/hnn-content/uploads/Pakistan20171.pdf>

of the facilities undertook the complete newborn assessment. None of the facilities had compiled information on neonatal deaths and stillbirths; recording of very low birth weight was missing in most (75%) of the facilities and most (77%) of the facilities did not receive feedback on out-referrals. The study also captured experiences and perceptions of caregivers' regarding financial burden during their stay in the facility. Around 90% of the caregivers reported spending out of pocket payments to bear expenses for infant care. 54% of the caregivers considered out of pocket payment expensive.

25. Experience and perceptions of caregivers' regarding financial burden during their stay in the facility.



Sick newborn care has been included in the list of services covered under the 'sehat sahat' card scheme. The direct medical costs are covered, however it is clear that other major costs of diapers, food and transport is substantial. Formula milk may not be required if mothers are allowed to stay in and expressed milk is encouraged.

Improving quality of care for small and sick newborns has shown significant impact on reducing Neonatal Mortality Rate in multiple settings⁴⁸. Having standardized recording and reporting tools, setting up quality of care teams and supporting them with training and enabling environment, perinatal death reviews and family-centred care are some of the successful strategies that can be adopted. Orientation of service providers on WHO's standards for small and sick newborn care⁴⁹ is in dire need.

Some efforts have been initiated for neonatal death reviews including sms reporting in Punjab based on LHW reports and testing the validity of the WHO revised verbal autopsy tool in determining causes of

⁴⁸ Nabila Zaka, et al., Quality improvement initiatives for hospitalised small and sick newborns in low- and middle-income countries: a systematic review, Implementation Science, 13, 1, (2018).

⁴⁹ Standards for improving quality of care for small and sick newborns in health facilities. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.

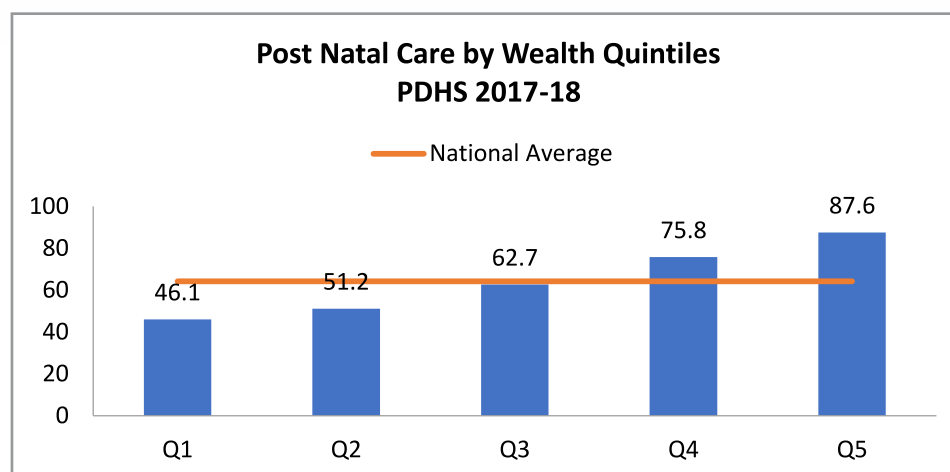
neonatal deaths⁵⁰. However perinatal death reviews, as part of the overall MPDSR system, is a huge gap in Pakistan.

Who is left behind?

The PSLM 2019 report highlights the disparities by household residence in Neonatal Mortality Rate. The NMR is 41/1000 live births nationally with 32/1000 live births for urban and 44/1000 live births for rural areas.

Global evidence suggests that about 62% of total neonatal deaths occurred during the first 3 days of life; the first day alone accounted for two-thirds⁵¹. Timely and quality PNC services are important for well-timed detection and management of any complications. Post Natal Care for babies is 64.2% nationally with wide variation (rural 57.7 versus urban 77.3). PNC coverage also varies with maternal education, with 50.3% for mothers with no education, 65% for primary education and 81.7% with secondary education and above. High difference is noted between the poorest and richest quintiles as shown in the graph. (DHS 2017-18)

26. Postnatal care by wealth quintiles



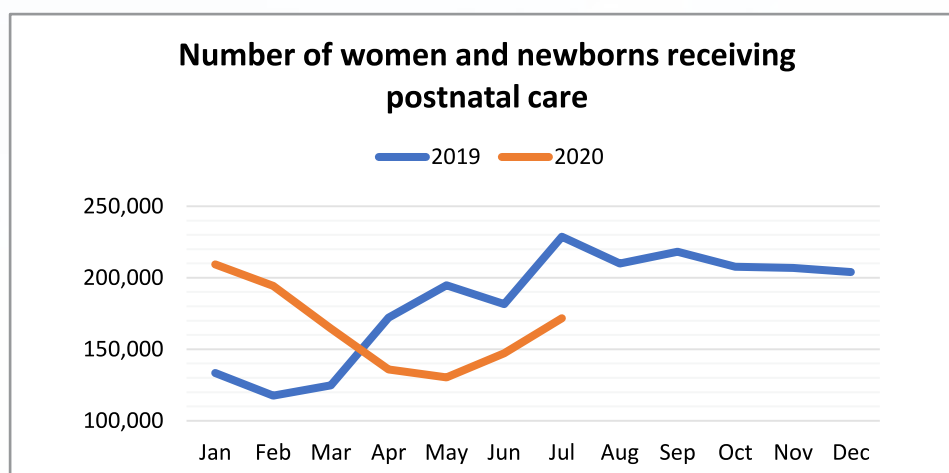
⁵⁰ Soofi, S.B., Ariff, S., Khan, U. *et al.* Diagnostic accuracy of WHO verbal autopsy tool for ascertaining causes of neonatal deaths in the urban setting of Pakistan: a hospital-based prospective study. *BMC Pediatr* **15**, 144 (2015). <https://doi.org/10.1186/s12887-015-0450-4>

⁵¹ Sankar, M., Natarajan, C., Das, R. *et al.* When do newborns die? A systematic review of timing of overall and cause-specific neonatal deaths in developing countries. *J Perinatol* **36**, S1–S11 (2016). <https://doi.org/10.1038/jp.2016.27>

COVID-19 Impact

Ministry of Health Services, Regulation and Coordination has documented a decline in PNC cases after lock down measures in April and May 2020. The number of PNC consultations in 2020 were significantly higher than 2019 but dropped during the months of April and May during extensive lockdown period.

27. Postnatal care consultations (DHIS data)



Strengthening Intensive Care Units for COVID-19 is an opportunity to strengthen neonatal needs for ICUs and safe oxygen. Similarly the system set-up for sentinel surveillance from public and private health facilities can also be sustained for running SAAVY (Sample Vital Statistics and Verbal Autopsy) for neonatal deaths.

CHAPTER 4: CHILD HEALTH

Table 14: Main points

	Progress	Challenges
Policy / context	In 2018, the Government of Pakistan made a commitment to reduce U5MR by less than 25 per 1000 live births by 2030	
Child mortality	Under-five mortality per 1,000 live births dropped steadily during the last 2 decades at an annual rate of decline just under 3% to 67 per 1,000 live births by 2019 The decline is faster in older infants and young children than neonatal deaths which now constitute nearly half of all under-5 deaths; Diarrhea and pneumonia are the leading causes of post neonatal and early child mortality.	Pakistan needs to accelerate its efforts to achieve SDG 3.2 by focusing on neonatal health and averting preventable child deaths due to diarrhea, pneumonia and other conditions. The average annual rate of decline needs to increase to about 8% per year.
Coverage	Access and utilization of services continued to improve for ARI and childhood diarrhoea. 70-80% of sick under-fives were taken to a health facility, and 53% of those with diarrhoea received ORS fluids in 2018/19 Coverage of fully immunized children has increased from 54 to 66% in 2017/18 and further increases in pentavalent third dose and measles vaccination were observed in the PSLM 2018/19	Urban-rural and provincial inequalities persist and the poorest wealth quintile have much lower coverage Quality of care is a challenge whereby nutritional assessment, age-appropriate vaccination, ORS plus zinc and continued feeding during diarrhea and provision of antibiotic for managing ARI could be improved by strengthening knowledge and counseling skills of service providers
Quality of care	ICT SARA survey found sub-optimal availability of essential medicines and commodities in public sector facilities	Recent decline in availability of medicines with LHWs is a grave concern. Health facility data also show chronic shortages of essential medicines. Appraisal of knowledge, attitude and practices of trained doctors regarding IMNCI revealed gaps
COVID-19	Immunization-related visits in June recovered and were at the same level as the year before	During April – May – June the number of children with ARI seen in facilities was down by 41% compared to the preceding year. The corresponding number of children with diarrhoea decreased by 39.7%. Immunization coverage was down by 34.2%.

Way forward	Reaching all children with effective interventions including immunization and treatment of sick children is essential. This implies well-stocked health facilities that are accessible and affordable to all, run by competent knowledgeable health workers. There is still considerable scope for improvement of the current public (and private services). Furthermore, under-5 mortality can only be brought down rapidly if neonatal mortality interventions are implemented effectively everywhere (see chapter on newborns). All interventions for under-fives should pay special attention to the children from the poorest households which still have lower coverage and higher mortality. Inter-sectoral linkages to combat malnutrition, improved household water supply, increased hand washing practices and reducing in-door air pollution are important for child health.
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Introduction

This chapter briefly reviews the main policies and targets for child health and describes main program implementation efforts. This chapter also presents the epidemiological context of under-five mortality trends in Pakistan.

The assessment of progress uses the cascade of effective coverage which aims to disentangle issues related to access, utilization, readiness and quality of services. Several steps in the cascade have a good level of available data from the population-based surveys (especially PDHS 2017/18 and regional MICS surveys), public sector health facility data, health facility assessments and research, allowing the identification of areas of progress and major challenges. Maternal and child nutrition is an important underlying risk factor; but is not included in this analysis and planned to be covered in the annual report for 2021.

This chapter assesses:

- level and trends of the care for pneumonia, diarrhoea and malaria
- access to services, readiness and quality of services
- extent to which services are received and used (immunization) and
- effective coverage for full immunization

Main Policies

By 2030, the SDG target (3.2) is that globally under-five mortality rate is reduced to 25 per 1000 live births or less. De facto, many countries have translated this into a national target. In 2018, the Government of Pakistan made a commitment to reduce U5MR by less than 25 per 1000 live births by 2030.

Program implementation efforts

In 1998, Pakistan piloted the Integrated Management of Childhood Illnesses (IMCI) strategy in two districts of Abbottabad and Multan. By 2012 all districts have started implementing the IMCI program. Neonatal care was added to 'IMCI' later making it 'IMNCI'.

This was further strengthened by training conducted under the National MNCH Programme. However, studies have highlighted a gap in understanding the management guidelines of IMNCI despite training⁵². Lady Health Workers also played a crucial role to perform their regular home visits and deliver promotive, preventive and simple but essential curative home health care services including the most common diseases encountered among the under 5 children population i.e. diarrhoea, respiratory tract infections and fever; by eagerly managing anaemia, early initiation of breastfeeding and childhood feeding, and immunization⁵³. Mother and Child Health Weeks⁵⁴ were conducted bi-annually to provide household awareness about child health, provide deworming tablets with refresher trainings to service providers at primary health care facility and conduct catch-up immunization starting from 2007. These campaigns still continue with names of child health and nutrition weeks in certain districts after the eighteenth amendment. An evaluation of mother-child weeks had shown significant improvements in coverage over-time and rated it as a doable strategy for providing a bundle of required interventions at periodic intervals, with low cost (Rs.43) per beneficiary. Recommendations were made to expand the focus to non-LHW areas for equity and invest in communication campaign for raising family awareness and community involvement⁵⁵.

Vaccinations for rota virus⁵⁶ and pneumococcal pneumonia have been added to the Expanded Programme for Immunization (EPI). A multi-center study in 5 developing countries including Pakistan revealed that Rotavirus was found to be the most frequently detected pathogen in diarrheal episodes, during the first year of life, with the highest incidence (20%) occurring among 6-11 months old⁵⁷. In addition, there are 15 million episodes of ARI every year among under-fives in Pakistan⁵⁸.

Pakistan also piloted the implementation of WHO's revised guidelines of Possible Suspected Bacterial Infection in Sindh⁵⁹ and has started scaling up in four Punjab since 2018.

A study showed that belonging to the lowest wealth quintile, lack of maternal education, previous birth interval < 2 years, first birth order and below average birth size were associated ($p < 0.05$) with a greater risk of child mortality in Pakistan. Common factors associated with child mortality were fever, diarrhoea and pneumonia⁶⁰.

⁵² Huma Amin ,Hafsa Yasin ,Syed Hasan Danish ,Farah Ahmad ,Shahida Rasheed ,Nosheen Zehra ,Gati Ara , Appraisal of knowledge, attitude and practices of trained doctors regarding IMNCI, Pak. J. Med. Dent. 2015;4 (1): 36-40

⁵³ Hafeez, A., Mohamud, B., Shiekh, M., Shah, S., Jooma, R. (2011). Lady health workers programme in Pakistan: challenges, achievements and the way forward.JPMA: Journal of the Pakistan Medical Association, 61(3), 210-215.

⁵⁴ <https://www.youtube.com/watch?v=dKApdH4wmzg>

⁵⁵ Assad Hafeez et al.(2012)The detailed assessment of Mother Child Week, Coverage Survey Report prepared for the Ministry of Health and NP for FP and PHC.

⁵⁶ <http://www.epi.gov.pk/vaccine-preventable-diseases/diarrhoea/>

⁵⁷ Ali, N. K., & Bhutta, Z. A. (2003). A review of rotavirus diarrhea in Pakistan: how much do we know?. Journal of the College of Physicians and Surgeons--Pakistan: JCPSP, 13(5), 297–301.

⁵⁸ <http://www.epi.gov.pk/vaccine-preventable-diseases/pneumonia/>

⁵⁹ <https://www.harponet.org/wp-content/uploads/2019/11/Evaluating-the-Scale-up-of-PSBI- WEBINAR-20-11-19.pdf>

⁶⁰ Afshan, K., Narjis, G., Qureshi, I.Z. and Cappello, M. (2020), Social determinants and causes of child mortality in Pakistan: Analysis of national demographic health surveys from 1990 to 2013. J Paediatr Child Health, 56:457-472. doi:10.1111/jpc.14670

Pakistan needs investments in strengthening implementation of policies designed to reduce child mortality amongst the most vulnerable. Strengthening birth and death registration and using sentinel sites for surveillance of child deaths are required to guide programmatic improvements.

General epidemiological trends: minimal decline in under-five mortality with elusive SDG target

The PDHS 2017-18 indicates a decline in under-five mortality to 74 per 1000 live births, from 89 in the PDHS 2006/07, both for the 3-year period preceding the survey. The under-5 mortality in PSLM 2018/19 was even lower (59 per 1,000 live births). The UN IGME considered this an underestimate of their estimated under-5 mortality in 2019 at 67 per 1,000 live births (with an average annual rate of reduction of 2.9% per year. These estimates imply an average annual rate of decrease of 8% per year to reach the 2030 target of 25 per 1000 live births, almost three times faster than the rate of decline in the last decade.

28. Trends in childhood mortality, 1990 to 2018

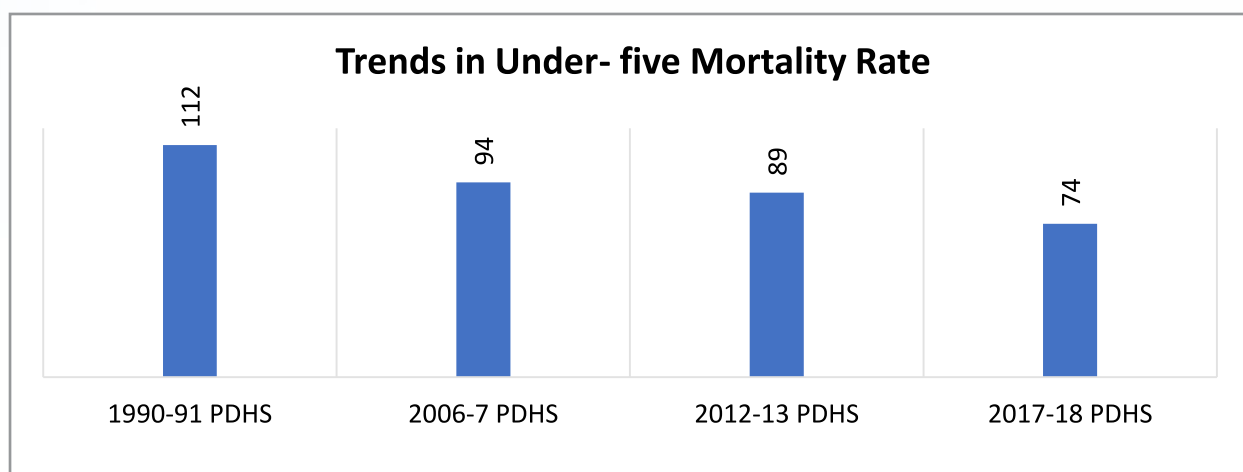
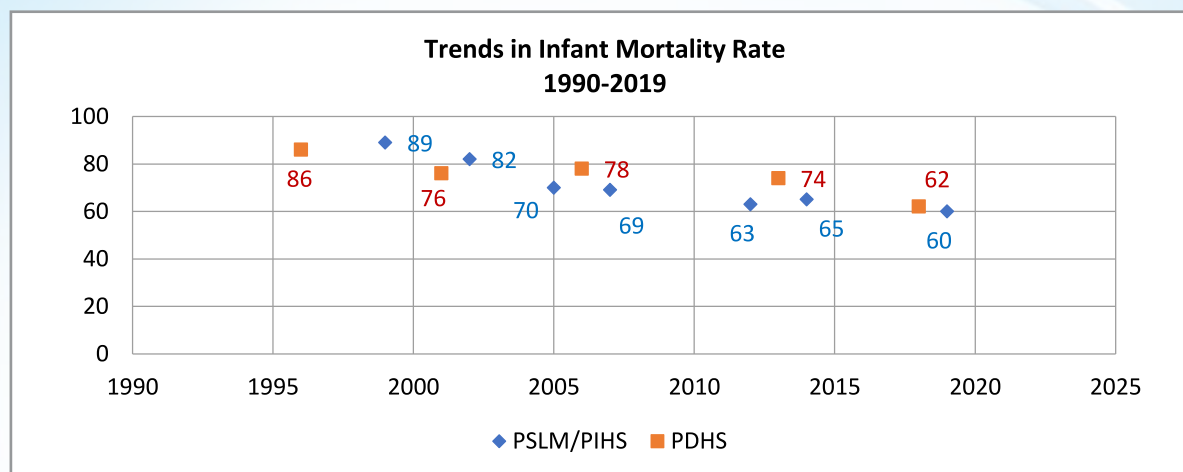


Table 15. Under-five mortality per 1000 live birth in the Pakistan DHS 2006-07 and Pakistan DHS 2017-18 with rates of change and required pace of decline to achieve the SDG target

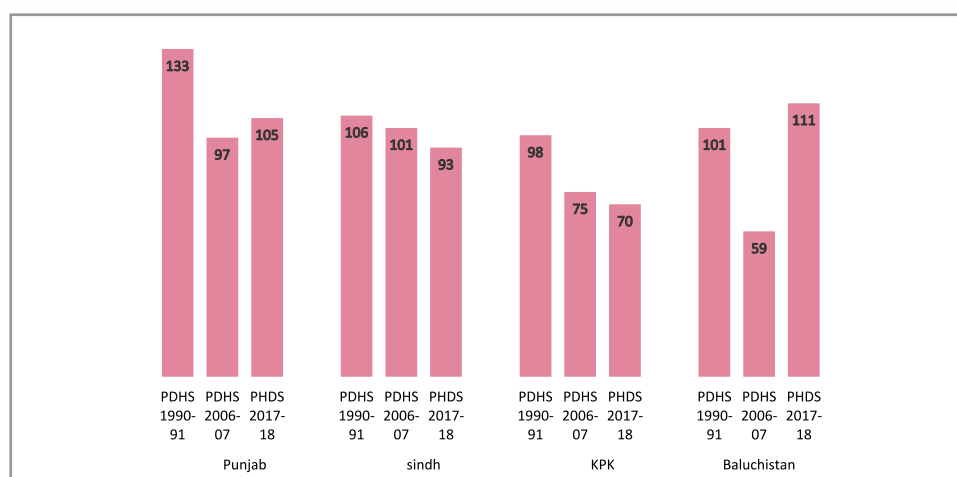
Survey	Mid-point of the 3 years preceding the survey	Under-five mortality per 1000 live births
PDHS 2006-07	2005.5	94
PDHS 2017-18	2018.2	74
PSLM	2016.5	59
Estimate UN IGME	2019.5	67
Average annual rate of change (UN IGME 2000-2019)		-2.1%
SDG	2030.0	<25
Average annual rate of change required		-8%

29. Trends in infant mortality rate 1990-2019



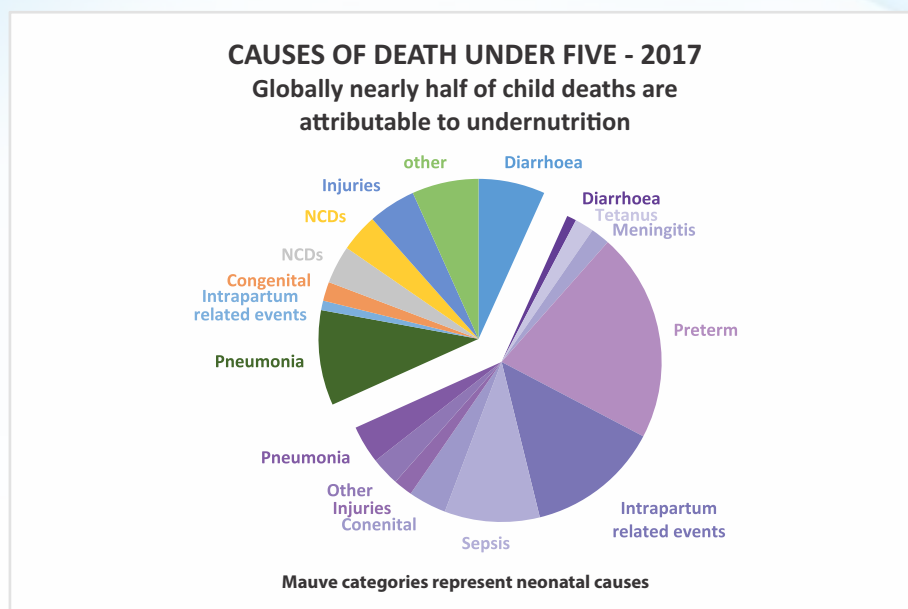
All provinces and regions with trend data have shown some progress in reducing U5MR since 1990.

30. Trends in Provincial Under-five Mortality Rates



In Pakistan, the majority of under-five deaths are attributed to neonatal complications of prematurity, intra-partum related events and sepsis. Other than these diarrhoea and pneumonia are the leading cause of death in children under five, according to global estimates that included the PDHS 2006/07.

31. Causes of under-five deaths



Source: <http://profiles.countdown2030.org/#/cp/PAK>

Surveillance for congenital anomalies has not yet been initiated at scale in Pakistan. The causes and patterns need to be understood for a focused action plan. Congenital anomalies can cause miscarriage, stillbirth, and for those who survive, lifelong physical and psychological difficulties. The lack of medical, surgical and support services to care for children affected, especially in low-resource settings, can take a toll on families and children⁶¹.

Assessment of progress

Access to services

Child health care services, preventative and curative including immunization are part and parcel of primary health care service provision in Pakistan. At community level, Lady Health Workers are trained in growth monitoring and IMNCI; treat mild cases and refer the sick children to health care facilities. The EPI programme runs the immunization services through the fixed centre and outreach by vaccinators. Polio vaccination campaigns are supported by LHWs, vaccinators and local volunteers. Recent declines in availability of medicines with LHWs is a grave concern. Health facility data also show chronic shortages of essential medicines. For example, ICT SARA survey 2020 shows that 66% of facilities (tertiary hospitals, CHCs, RHCs, FWCs, BHUs and dispensaries) were providing preventative and curative services for children under-five. The FWCs and dispensaries were particularly low on availability of child health services.

⁶¹ WHO, CDC, March of Dimes (2020) Birth defects surveillance: a manual for programme managers and Birth defects surveillance: quick reference handbook of selected congenital anomalies.

Table 16. Access to services in ICT

Preventive and curative care for children under 5	Malnutrition Diagnostics and treatment	Vitamin A supplementation	Iron supplementation	ORS and Zinc supplementation	Growth monitoring	Treatment of pneumonia	Administration of amoxicillin for the treatment of pneumonia in children	Treatment of malaria in children
66%	48%	19%	31%	23%	33%	33%	42%	30%

The readiness scores for medicines and equipment were also un-satisfactory in ICT SARA survey.

Table 17. Access to services in ICT

Guidelines and trained staff					Equipment				
Guidelines for growth monitoring	Staff trained in IMCI	Staff trained in growth monitoring	Guidelines and trained staff readiness score	Child and infant scale	Length/height measuring equipment	Thermometer	Stethoscope	Growth chart	Equipment readiness score
0%	6%	3%	2	16%	21%	78%	95%	10%	44

An appraisal of Knowledge, Attitude and Practices of trained doctors regarding IMNCI revealed gaps in knowledge and practices of trained doctors⁶² regarding management according to IMNCI guidelines. This is also substantiated by recent ICT SARA survey findings as shown above.

Innovations like on-line courses on IMNCI for doctors are being successfully tried in Punjab and can reduce the training costs significantly. Inclusion of IMNCI and growth monitoring to pre-service curricula is also recommended.

Similar gaps were also found in ICT SARA survey for immunization related guidelines, staff capacities and equipment.

⁶² Amin H, Yasin H, Danish, SH, Ahmad F, Rasheed S, Zehra N, Ara G. Appraisal of knowledge, attitudes and practices of trained doctors regarding IMNCI. Pak J Med Dent 2015; 4(1):47-53.

Table 18. Immunization related guidelines in ICT

Guidelines and trained staff			Equipment							
Guidelines for child immunization	Staff trained in child immunization	Guidelines and trained staff readiness score	Cold box/vaccine carrier with ice packs	Refrigerator	Sharps container/safety box	Auto-disable syringes	Temperature monitoring device in refrigerator	Adequate refrigerator temperature	Immunization cards	Immunization tally sheets
14%	21%	17	21%	28%	31%	23%	27%	8%	23%	12%

Improving access through engagement of the private sector is important. The private sector caters to a substantial need for child health care except immunization. General Physicians and pharmacies are providers for treatment of common childhood illnesses. PDHS 2017-18 documented that a private hospital or clinic was the source of advice or treatment for more than half of the children (52%).

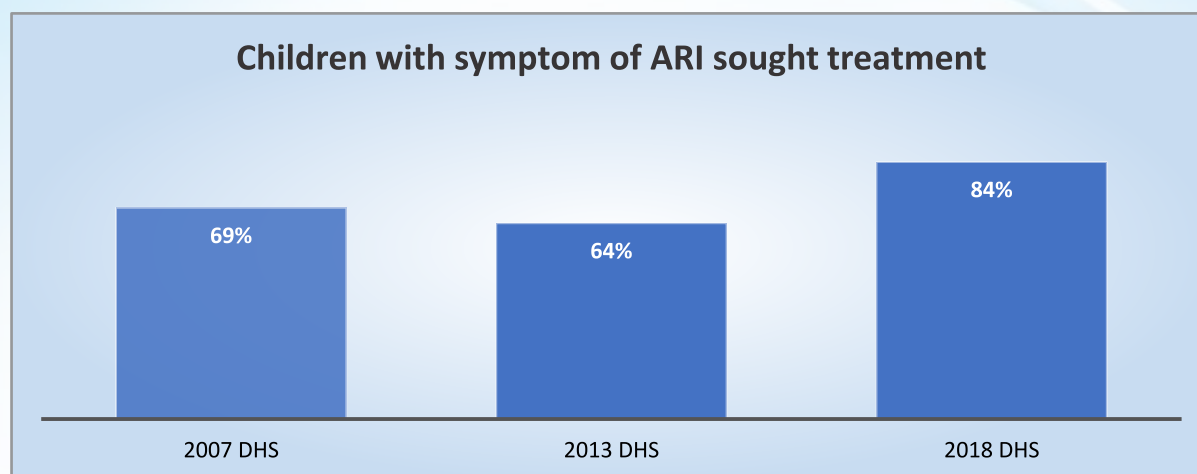
Utilization of Services / Contact Coverage

Care seeking for pneumonia

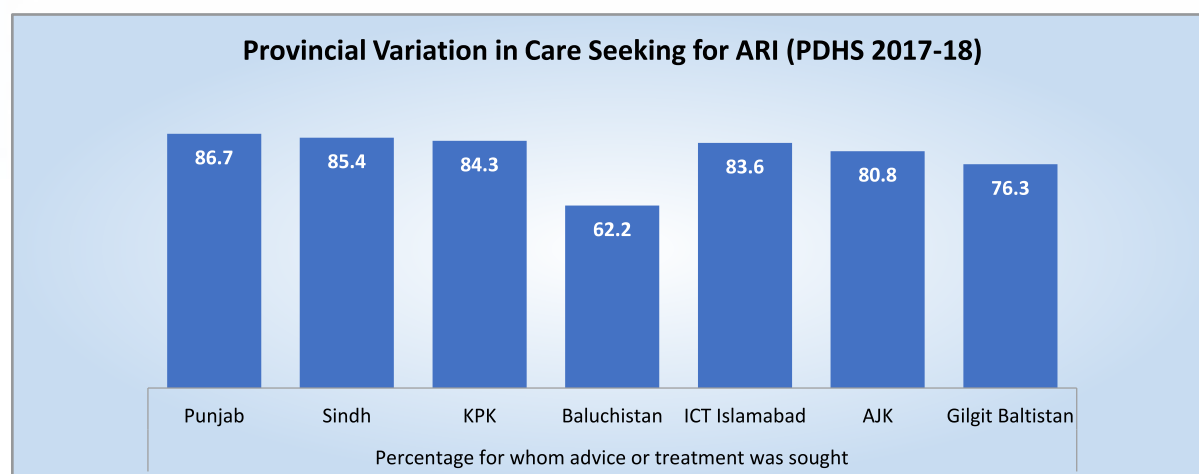
PDHS 2017-18 showed that 14% of children under age 5 showed symptoms of acute respiratory infection (ARI), 84% were taken to a health facility or care provider for advice or treatment, 51% sought treatment in the same or next day, and 46% were given antibiotics as treatment.

Incidence of ARI has declined from 16% in 2012-13 to 14% in 2017-18. The prevalence of symptoms of ARI was highest among children age 6-11 months and age 12-23 months (17%) and lowest among children 48-59 months (10%).

32. Percent of children under 5 years with symptoms of pneumonia taken to appropriate health provider



33. Percent of children under 5 years with symptoms of pneumonia taken to appropriate health provider



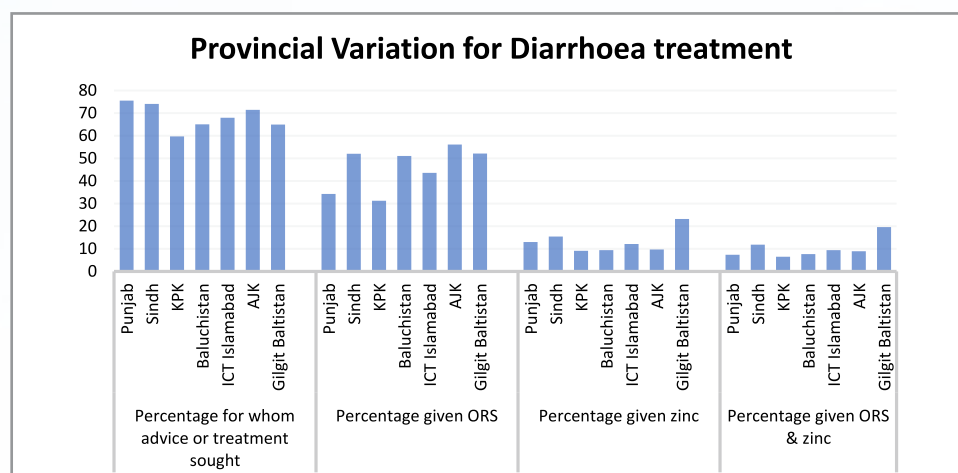
Children in households from the lowest wealth quintile are least likely to receive immediate treatment for ARI (36%), receiving treatment on the same or next day, compared with 68% of children in the highest wealth quintile. Children in Balochistan with symptoms of ARI are least likely to seek advice or treatment (62%) compared with those in Punjab (86%), Sindh (85%) and Khyber Pakhtun Khwa and ICT (84% each). They are also least likely to receive antibiotic drugs.

Care seeking for diarrhea

There is a downward trend in the prevalence of diarrhoea in children under age 5 from 23% in the 2012-13 PDHS to 19% in the 2017-18 PDHS. The practice of seeking advice or treatment for diarrhoea improved from 61% in 2012-13 to 71% in 2017-18. In the PSLM 2018/19 diarrhoea in the last 15 days was reported for only 6% of children under-5. Among those, 84% were taken to a health facility and 53% received ORS.

Only 37% of the children with diarrhoea were given oral rehydration salts, and only 8% were given zinc with ORS. Appropriate stool disposal and sanitation is important for prevention of diarrhoea. Only 36% of children's stools were disposed of appropriately. (PDHS 2017-18)

34. Provincial trends for treatment of diarrhoea among children under-five



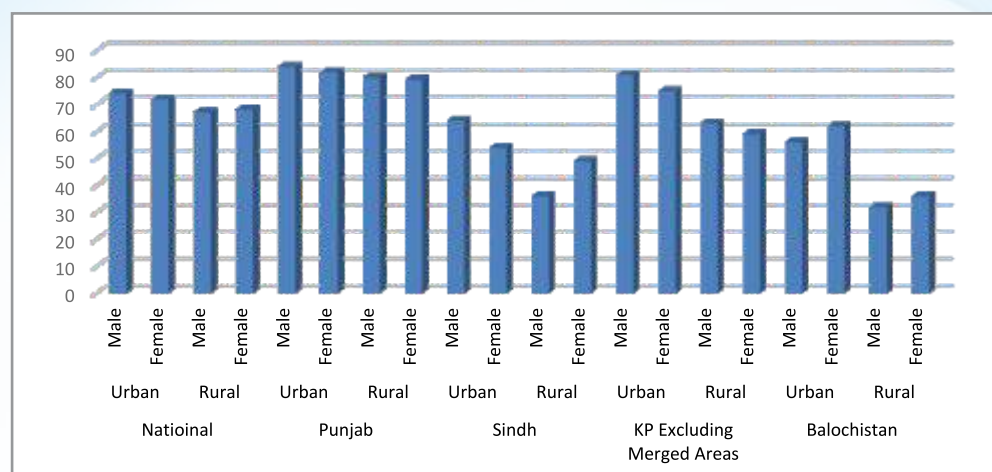
Co-packaging of ORS and Zinc and the “Diarrhea Pack” are two promising interventions to increase the use of zinc. The Diarrhoea Pack comprises of the two packets of low osmolality ORS, one strip of 10 zinc tablets, two packets of water purification sachets and one pictorial leaflet with educational material including importance of hand washing, use of toilet facilities and safe storage of water and food, distributed through community health workers. A study⁶³ assessing the use of the Diarrhoea Pack has shown a significant increase in uptake of ORS and Zinc along with the reduction in antibiotic use, diarrhea burden and hospitalization in intervention clusters when compared with the control clusters.

Immunization Services

The percentage of fully immunised children ages 12-23 months has increased markedly over a 5-year span, from 54% in 2012-13 to 66% in 2017-18. The PSLM 2018/19 shows coverage by vaccine for children aged 12-23 months. Pentavalent third dose and measles coverage were 85% and 77% respectively. This compares to 75% and 73% respectively for the year before in the PDHS 2017/18. Although the surveys may have design or implementation differences, these data suggest that immunization coverage is still increasing.

⁶³ Habib et al. 2013. A study to evaluate the acceptability, feasibility and impact of packaged interventions (“Diarrhea Pack”) for prevention and treatment of childhood diarrhea in rural Pakistan BMC Public Health 2013, 13:922 <http://www.biomedcentral.com/1471-2458/13/922>

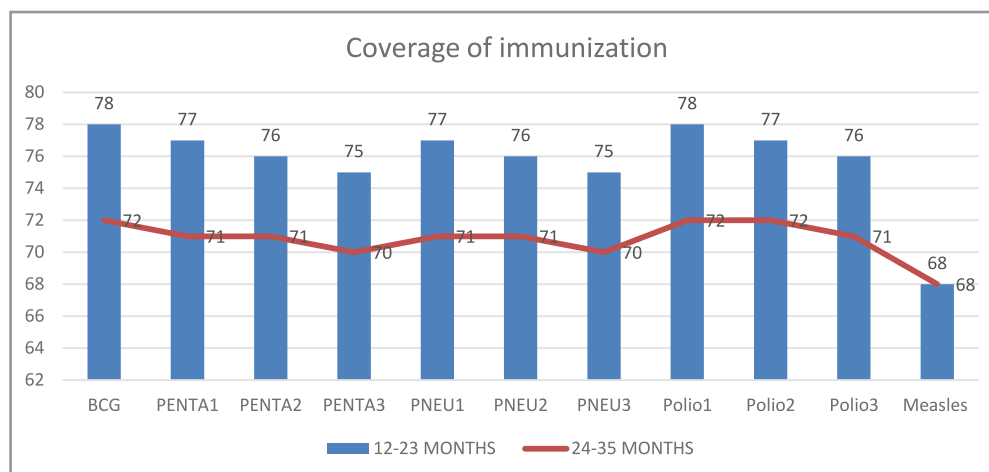
35. Percentage of children age 12-23 months vaccinated at any time before the survey



Source: PSLM 2018-19

Vaccination card ownership and availability is important as a card ensures that children receive all the recommended vaccinations. Eighty five percent of children aged 12-13 months were reported to have a vaccination card; however only 63% of children's vaccination cards could be seen at the time of the interview. Plastic coated EPI books are now being used to address this issue.

Figure 36: Coverage of immunization



PSLM 2019 shows an improvement in national coverage for all antigens (as shown above in Fig 36). Measles coverage has been stagnant and drops due to a time gap and possible omission by parents after Penta 3.

Readiness of services: stock outs of pneumonia and diarrhoea commodities and missed opportunities for immunization

ICT SARA survey found sub-optimal availability of essential medicines and commodities in public sector facilities. It is note-worthy that co-packaged ORS and Zinc is not yet a part of the Essential Medicine List.

Table 19. stock outs of pneumonia and diarrhoea commodities in ICT

Medicines and commodities						
Oral rehydration solution packet	Amoxicillin (dispersible tablet 250 or 500 mg OR syrup/suspension)	Co-trimoxazole syrup/suspension	Paracetamol syrup/suspension	Vitamin A capsules	Me-/albendazole cap/tab	Zinc sulphate tablets, dispersible tablets or syrup
59%	55%	9%	71%	2%	40%	16%

The same survey had the following findings for the availability of immunization services:

Table 20. Missed opportunities for immunization in ICT

Location/place/room of immunization services easily recognized by clients	Location/place/room of immunization services comfortable for vaccinators and clients	Waiting area for the clients	Comprehensive EPI micro-plan	RED/REC microplan for union council
27%	24%	30%	15%	9%

Availability of vaccines was lowest at FWCs and dispensaries. The facilities reporting stock-outs for all the vaccines in the last 3 months varied from 3-5%. HPV is not part of Pakistan's EPI programme at the moment.

Table 21. Vaccine stock out in ICT

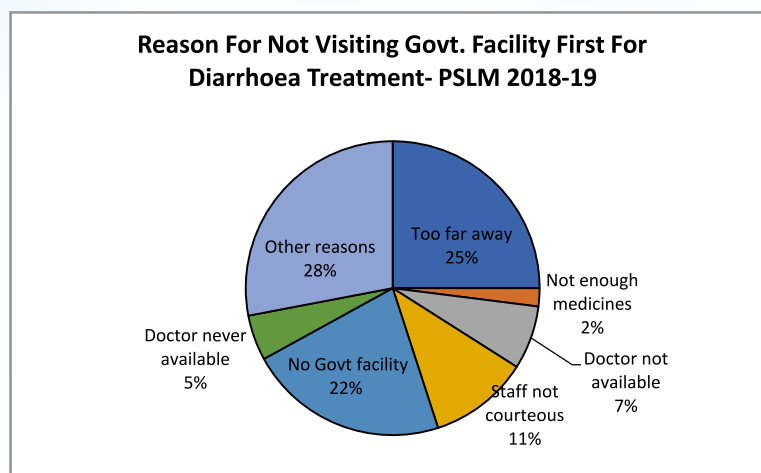
Vaccine							
Stock out Measles vaccine	Stock out DPT-Hib+HepB vaccine	Stock out Oral polio vaccine	Stock out BCG vaccine	Stock out Pneumococcal vaccine	Stock out Rotavirus vaccine	Stock out IPV (Inactivated Poliovirus Vaccine)	Stock out HPV (Human Papillomavirus)
5%	3%	3%	5%	5%	3%	5%	n/a

Quality of care: issues of prioritization and measurement

For the purpose of this document, combined use of ORS and zinc, use of antibiotic for treatment of pneumonia and age appropriate vaccination are reviewed as measures of quality child health care.

Although the practice of seeking advice or treatment for diarrhoea in children has increased from 61% in the 2012-13 PDHS to 71% in 2017-18 PDHS; many gaps in quality of care have been observed. It is highly recommended that ORS and zinc be given to children with diarrhoea to reduce the risk of childhood morbidity and mortality. According to the PDHS 2017-18, about 37% of children with diarrhoea in the last 2 weeks were given ORS and only 1 in 4 (8%) children received zinc with ORS. Only 9% of children who had diarrhoea were given more liquids than usual for rapid rehydration, while 55% were given the same amount as usual and 35% were given somewhat less or much less than usual. Thirty percent of children with diarrhoea received unnecessary antibiotics and 21% received anti-motility drugs. Safe disposal of child faeces is only practiced in 36% of households. Knowledge and counselling skills of service providers need improvement to adhere to evidence-based practices and encourage care-givers to improve quality of early care seeking and appropriate management.

37. Reasons for not visiting Govt. facility first for diarrhoea treatment- PSLM 2018-19



Care was sought for seventy six percent of children with Acute Respiratory Infection (ARI) but timely care-seeking on the same or next day was reported by only 35.6%. Out of 76% children for whom any care was sought, only 50.6% of children received antibiotic for management of ARI.

Overall 66% of children age 12-23 months have had all basic vaccines. Childhood vaccinations should be specific to the age of the child. Only 51% of children had all age-appropriate vaccinations. Only 4% did not have any type of vaccine at any time before the survey.

The WHO Standards for improving quality of care for children and young adolescents aged 0–15 years in health facilities⁶⁴ have been framed in the best interest of children and take into consideration a child's right to health while recognizing that their health, physical, psychosocial, developmental and communication needs differ from those of adults. Evidence based care is required with competent service providers, necessary triaging, nutrition assessment, well equipped facilities, referral support, updated patient records and in line with fulfilment of human rights. The gaps in service delivery have been highlighted in the sections above. Examples exist from countries showing post neonatal mortality reduction by improving quality of care⁶⁵. No systematic effort is yet in place to measure and implement standards of care for children and young adolescents.

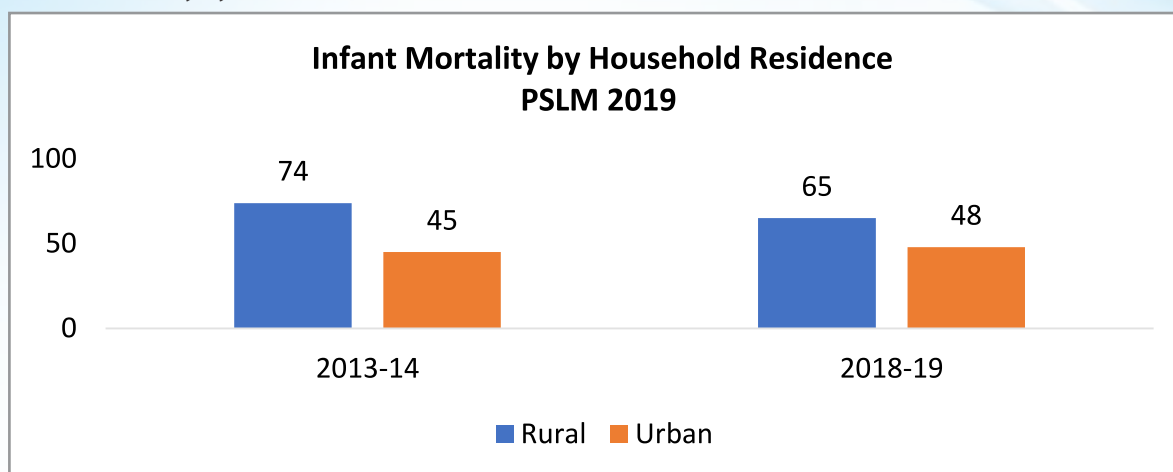
Who is left behind?

Urban rural disparity and differentials in Infant Mortality Rates by education status of the mother were noted in the recent PSLM 2019.

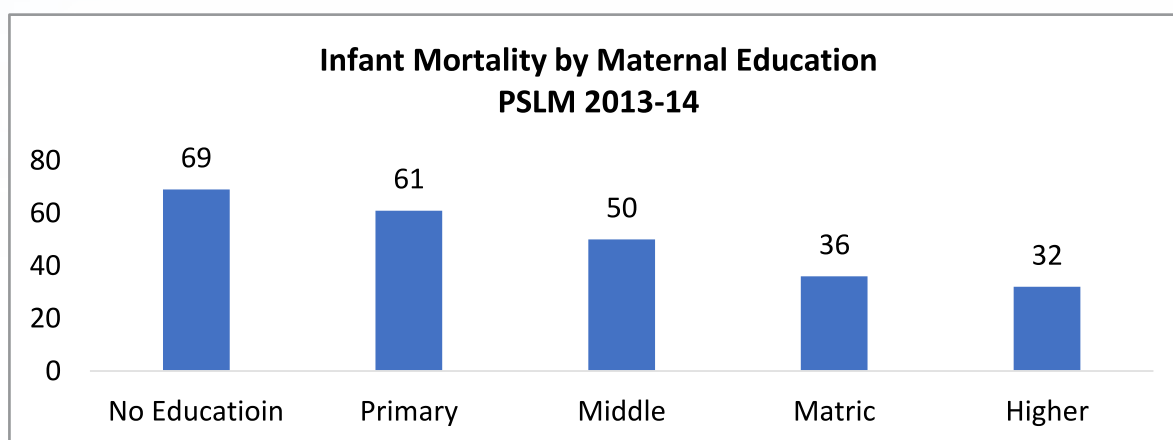
⁶⁴ Standards for improving the quality of care for children and young adolescents in health facilities. Geneva: World Health Organization; 2018. Licence: CC BY-NC-SA 3.0 IGO.

⁶⁵ Kavita Singh, et al., Evaluating the impact of a hospital scale-up phase of a quality improvement intervention in Ghana on mortality for children under five, International Journal for Quality in Health Care, Volume 31, Issue 10, December 2019, Pages 752–758, <https://doi.org/10.1093/intqhc/mzz073>

38. Infant mortality by household residence PSLM

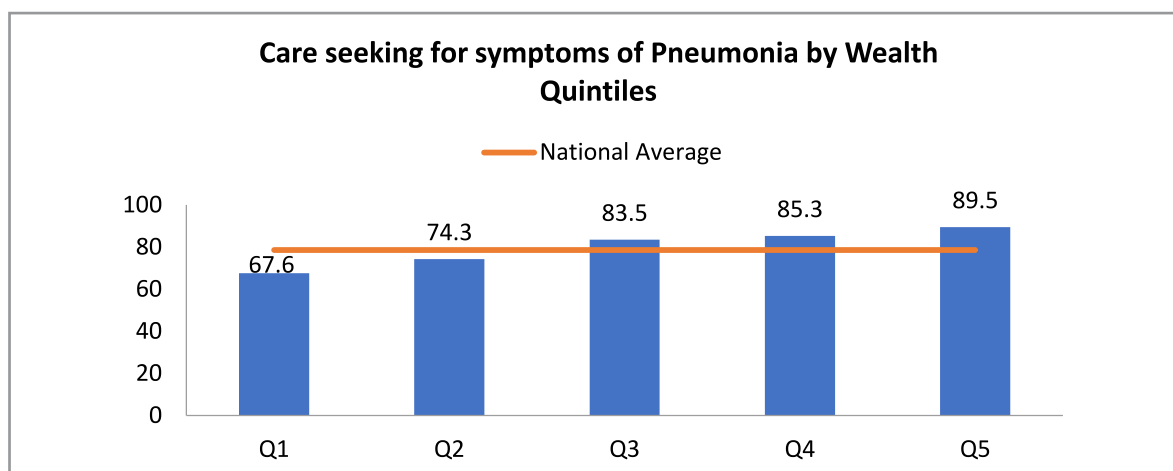


39. Infant mortality by maternal education PSLM 2013-14

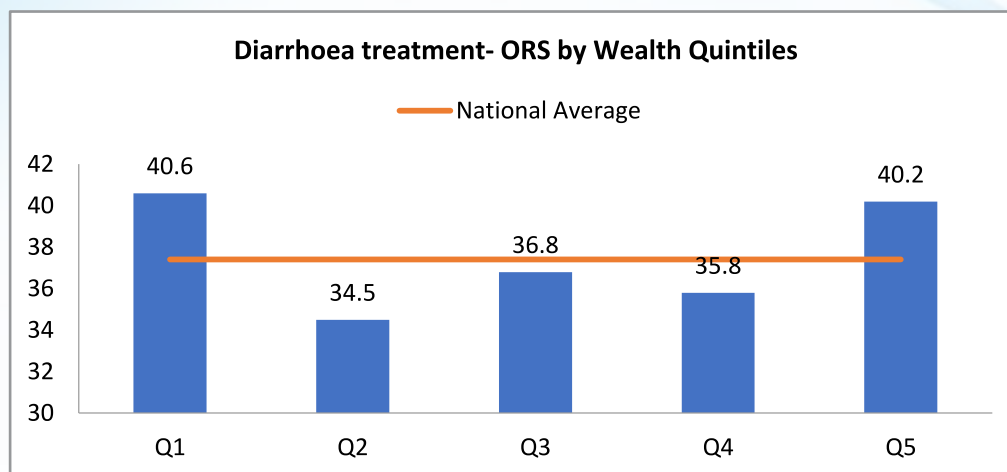


The variance in care seeking by wealth quintiles was also noted for pneumonia care seeking, diarrhoea treatment (use of ORS) and DPT3 coverage in PDHS 2017-18.

40. Care seeking for symptoms of pneumonia by wealth quintiles

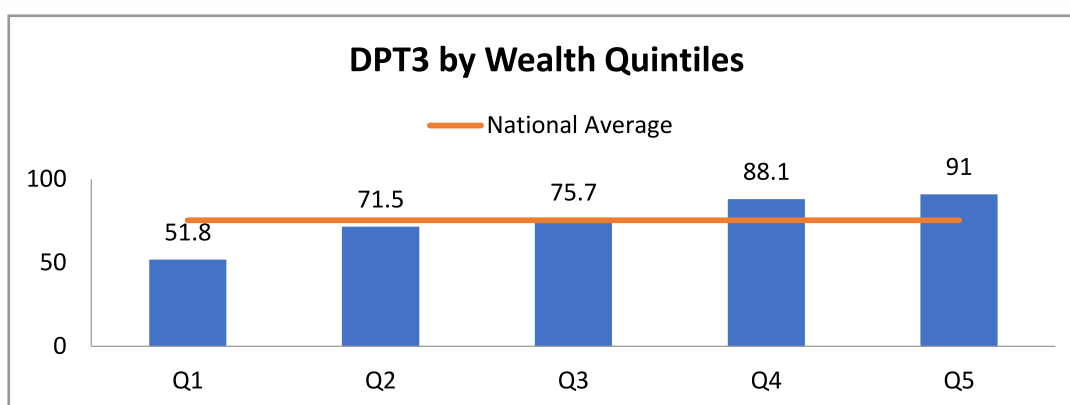


41. Diarrhoea treatment- ORS by wealth quintiles



It is interesting to note that while care seeking for pneumonia is 21.9 points lower among the poorest as compared to the richest quintile, use of ORS is similar among the highest and lowest quintiles with about a five point reduction for the remaining income groups. This pattern needs further exploration as to the patterns of non-financial causes on low ORS use.

42. DPT3 by wealth quintiles

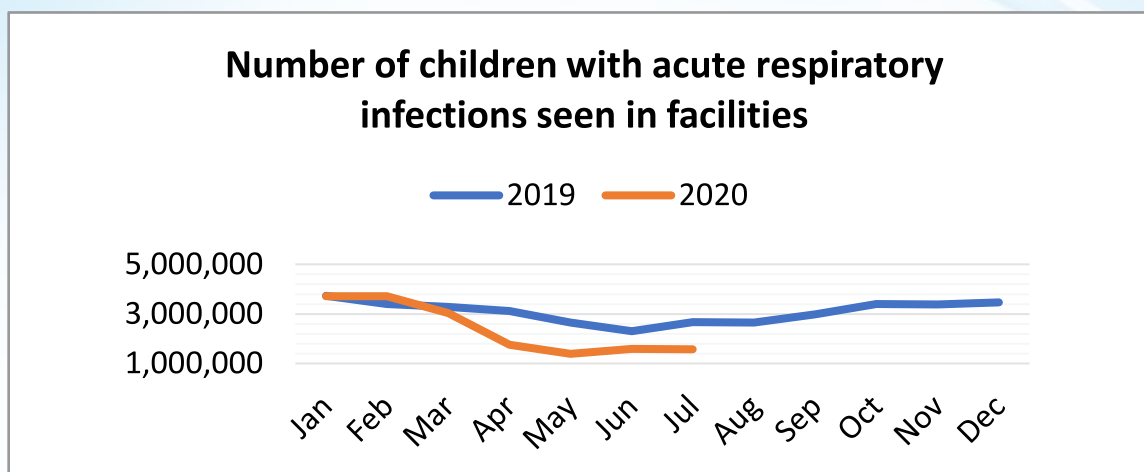


Children living in the poorest households are disproportionately disadvantaged for DPT3 coverage and need a targeted approach for service delivery.

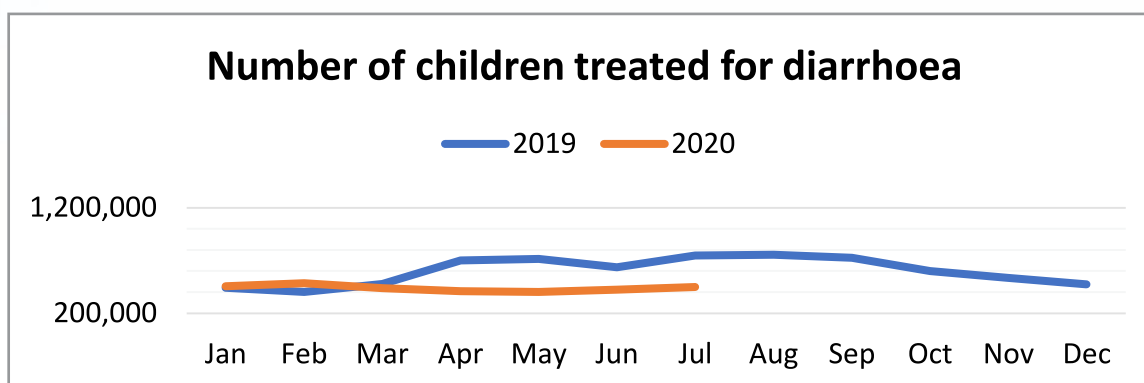
COVID-19 Impact

COVID-19 containment measures such as lockdown and closure of some facilities, had a negative impact on utilization of child health services at public facilities. DHIS reports compared for 2019 and 2020 clearly show the declining trends of consultations at government facilities. During April – May – June the number of children with ARI seen in facilities was down by 41% compared to the preceding year. The corresponding number of decrease in diarrhoea consultations was 39.7%.

43. Number of children with acute respiratory infection seen in facilities

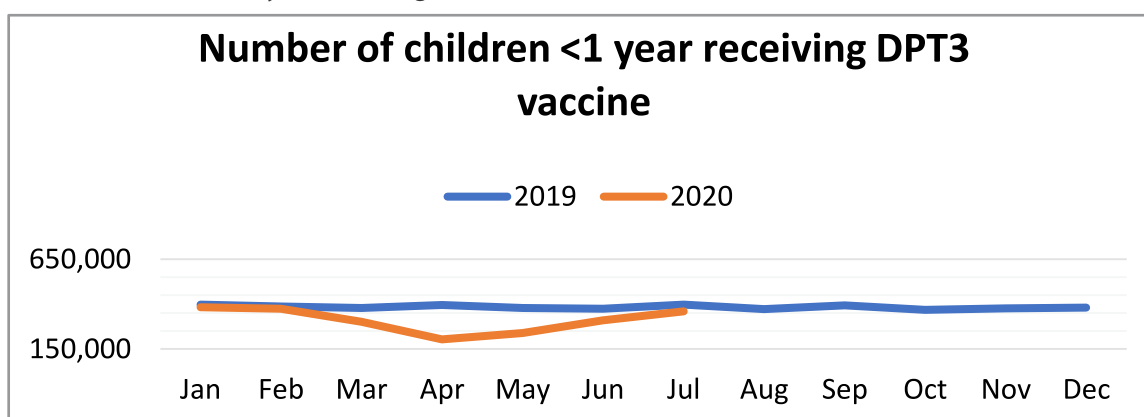


44. Number of children treated for diarrhoea



A similar decline in DPT3 vaccination was also registered during March to May 2020 with more drastic lockdown measures. However the service quickly resumed with advocacy efforts of immunization programmes. The decline in vaccinations was large (34.2% decline) during April-June but almost nil during July.

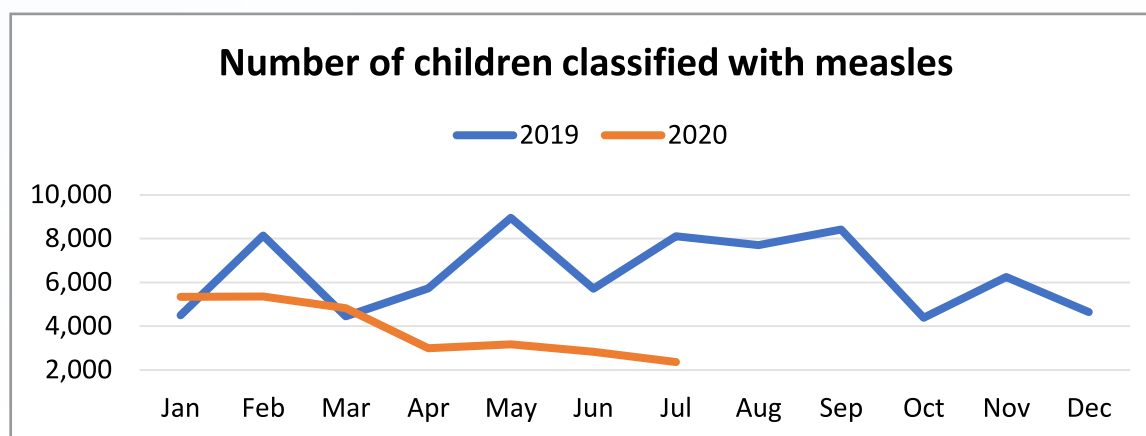
45. Number of children <1 year receiving DPT3 vaccine



A study⁶⁶ in Sindh province also showed a 52.5% decline in the daily average total number of vaccinations administered during lockdown compared to baseline, especially at the fixed sites. One out of every two children in Sindh province had missed their routine vaccinations during the provincial COVID-19 lockdown.

The number of measles cases was down during April to July 2020 as compared to 2019. This may be associated with reduced transmission because of the containment measures and with the lower chance of women taking their sick children to a health facility.

46. Number of children classified with measles



The Ministry of National Health Services, Regulations and Coordination has taken major steps to reinvigorate the immunization program to mitigate the impact of COVID-19. Low coverage areas were identified to accelerate outreach activities to cover children out of pocket, strengthen existing EPI centres and integrate of EPI and PEI (Polio Eradication Initiative) strategies for increasing equitable coverage of immunization against vaccine preventable diseases (VPD)⁶⁷.

By September 30th 2020, 4,634 children under-five were reported to be positive for COVID-19 in Pakistan. Guidelines for COVID management in children were developed and launched in July 2020⁶⁸ through collaborative efforts of the UNICEF and Pakistan Pediatric Association, under the leadership of MoNHSR&C.

⁶⁶ Subhash Chandir et al. 2020. Impact of COVID-19 pandemic response on uptake of routine immunizations in Sindh, Pakistan: An analysis of provincial electronic immunization registry data, Vaccine, Volume 38, Issue 45, 2020, Pages 7146-7155, ISSN 0264-410X, <https://doi.org/10.1016/j.vaccine.2020.08.019>.

⁶⁷ <http://www.epi.gov.pk/wp-content/uploads/2020/04/Federal-EPI-Updates-April-2020-1.pdf>

⁶⁸ <https://www.app.com.pk/national/unicef-launches-guideline-for-covid-19-management-in-children/>

CHAPTER 5: CALL FOR ACTION

Accelerating the progress on family planning, maternal, newborn and child health in a decade demands a de novo approach, high political commitment backed by adequate financing and the will to make difficult decisions for addressing redundancies and gaining efficiency. The recent developments in the RMNCAH+N investment case, UHC Basic Package and renewed Government commitment to invest in health have given an impetus to think bolder and bigger. A few key priorities for action are discussed in this chapter.

Tracking Effective Quality Coverage and Health Gains

The previous chapters have examined progress for family planning and maternal, newborn and child health using a six-step coverage framework, starting from a target population to (1) service contact, (2) likelihood of services, (3) crude coverage, (4) quality-adjusted coverage, (5) user-adherence-adjusted coverage and (6) outcome-adjusted coverage. Clear inferences can be made only if all the subsequent coverage levels are nested in the preceding ones and population denominator remains constant. For these purposes, PDHS 2017-18 is used as it allows some measure of quality care as well as highlights disparities by province and region, age, urban or rural residence, socio-economic status, sex of the child and mother's educational status. PSLM 2019 offers relatively recent data but does not cover regions (ICT, AJK and GB) and disaggregation variables are limited. Recent MICS data, though more elaborate on content of care, is not available for all provinces/regions. The sample sizes and design of each survey is also different which limits the comparability of information. The One Health Survey has been recommended in the past to overcome these limitations and should be seriously considered for efficiently managing the data needs.

Health facility assessments are useful to gauge facility readiness. Punjab, Sindh and ICT have used SARA survey for assessing public health facilities. The findings from the most recent ICT SARA survey were used to illustrate the gaps in commodities/medicines, staffing and availability of services in this report. DHIS reports have the potential of providing real time information from public health facilities. For example, the effect of COVID-19 on health facility usage through DHIS information has enabled the decision-makers to take timely measures for continuation of essential RMNCAH services. On the other hand, we could not assemble evidence on how the information on staff vacancies, stock-outs and maternal and U5 death is being used. DHIS does not report on causes of newborn deaths nor does it include reporting of still births (antepartum or intra-partum). Data from THQ hospitals and teaching hospitals is not included or linked to DHIS. Any future endeavours in this regard should consider inter-operability.

The private sector caters to almost 70% of all health care service delivery in Pakistan but no system exists to capture their information. Health Care Commissions in Punjab, Sindh and KPK are evaluating private facilities for meeting Minimum Service Delivery Standards. This information is confidential for individual facilities but aggregates can be released periodically to assess progress. There is as yet no provision of capacity building support for public and private health facilities for quality improvement. Private

hospitals are included in the list of empaneled facilities for using the 'sehat sahulat' card; and have shared the information on COVID-19 cases and number of available beds and ventilators. Partnerships with the private sector also exist to notify Dengue fever and TB cases. These examples show promise of bringing private health establishments into the health information system and simplified recording and reporting forms can be introduced through a participatory approach with GP associations.

Data availability for most of the SDG and UHC related indicators is reasonably adequate in Pakistan. However, huge gaps exist for CRVS (Civil Registration and Vital Statistics); birth weight measurement, reporting of still births and standardized measures for determining quality of care and cohort tracking. Extensive investment is required to scale-up and streamline MPDSR and link these efforts with continuous quality improvement.

Strengthening the Health System

A quality health system is needed to deliver the FP and MNCH interventions for achieving the desired health outcomes. At the current rate of progress, SDG targets for maternal, newborn and child mortality will remain elusive. There are no magic bullet answers for mammoth health system challenges. With the eighteenth amendment in place and health being a provincial responsibility, the departments of health have a pivotal role to play. Provincial lead on health is a great opportunity to situate health in the political priorities of the government and make the improvements at a quicker pace with attention to inter-district disparities and variations. This however runs the risk of fragmentation and lack of a unified approach which creates efficiencies. The National Ministry of Health Services Coordination and Regulation has successfully provided the platform of RMNCAH Technical Working Group which can be further strengthened.

Governance and leadership, especially at the provincial and district levels are key to design and deliver results. Under the current scenario, none of the provinces have the position of Director MCH. The Provincial MNCH programme units are on the verge of extinction as soon the extended PC-1s expire with exception of Sindh province where a Deputy DG RMNCAH position has been announced. Whether new positions for a technical team to oversee FP and MNCH are created or not will determine the provincial capacity to run an accelerated initiative to reach SDG targets. It may be practical to house this expertise in units under consideration for UHC basic package implementation for sustainability. The needs for technical assistance should be channelled through the National RMNCAH Technical Working Group for effective management.

The existing provincial Essential Packages for community based and primary health care and hospital services need a review to align with the newly endorsed UHC BP of services. The implementation of UHC offers the prospect of registration of families and creating unique IDs of patients for electronic medical records. Thus the monitoring and measurement efforts for cohort tracking will be possible to identify lapses in the effective quality coverage.

Service delivery improvements are dependent on resolving the serious health work force crises. Both pre-service and in-service trainings should be need based and cater to the competency gaps. Staff retention and motivation issues also need to be addressed. Referral pathways from district hospitals to teaching hospitals should be explicitly outlined with each teaching hospital linked to a few feeder district hospitals. This will create a rapport and enable teaching hospitals to take mentoring role for the teams at district hospitals. The models of pooled ambulance service have shown promising results and can be replicated in all districts.

Most of the MNCH and FP commodities are already included in the essential medicine list with exception of some brands of contraceptive implants and co-packaged ORS and Zinc. Procurement processes need clear SOPs and unified guidelines. Logistic Management and Information System (LMIS) strengthening is urgently needed to avoid stock outs.

Quality of care standards needs implementation for Family Planning, MNH and pediatric care with clinical mentoring guidelines and periodic third party assessments. Provincial health care commissions can support these assessments. However the capacity of public health managers and clinicians also needs to be built and an enabling environment ensured. Patients should be partners in health care, empowered and regularly consulted to evaluate their satisfaction and compliance.

Information systems inclusive of digital innovations are indispensable for timely and evidence-based decision making. COVID-19 surveillance measures have created a unique opportunity to integrate maternal, newborn and child death reporting in sentinel surveillance. Maternal Perinatal Death Surveillance and Response is a key to identify problems in patient management and health system. Linking these problems with quality improvement processes can guarantee incremental improvements and prevent unnecessary mortality. The newly created Integrated Disease Surveillance and Response units at district level can oversee the MPDSR activities and provide support to the health facilities.

Addressing the Underlying Causes of Preventable Mortality

The underlying malnutrition, air pollution and inadequate WASH behaviours drive maternal and child mortality, decreases resistance to environmental insults and increases the risk of respiratory infections and diarrhoea. Addressing these causes need inter-sectoral collaboration and a leadership by the health sector in advocacy and building a coherent response. The low level of human development, coupled with security, poverty, socio-economic challenges, health emergencies and recurrent humanitarian crises all contribute to vulnerabilities for women and children. MNCH and family planning need to be positioned at the centre of all development and risk preparedness efforts.

Leaving No One Behind

The low level of human development, coupled with security, poverty, socio-economic challenges, health emergencies and recurrent humanitarian crises are some of the macro level challenges resulting in

perpetual inequities in health care provision. At meso level, inadequate health financing, weak leadership capacities at district level, multiple parallel and fragmented health care structures and serious crises of the health workforce has resulted in a weak health system. At micro-level, low literacy, malnutrition and poor social status of women hamper proactive and timely health care seeking.

The concept of Universal Health Care (UHC) implies that all people and communities can use the promotive, preventive, curative, rehabilitative and palliative health services they need, of sufficient quality to be effective, while also ensuring that the use of these services does not expose the user to financial hardship. Access to essential health services and health equity are the key challenges in a country of 224 million people where an estimated 46 million people are living below the poverty line. There has been a consistent and significant decline in poverty over the years. The poverty head count measure fell from 64.3% in 2001 to 24.3% in 2015. However, since 2015, the pace of poverty reduction is likely to have halted due to macroeconomic crisis and associated slow-down in growth. In this context, the outbreak of the recent COVID-19 pandemic and the containment measures adopted by the government, are expected to increase poverty in the country. Less than one percent of the Gross Domestic Product (GDP) is being spent as governmental health expenditure and share of out of pocket health expenditure in Pakistan is around 57.6% of the total health expenditure, which is very high. In addition to the existing challenges of access to essential health services and health equity, the COVID-19 pandemic has intensified the need for investment in development of an equitable and resilient health system and mechanism for financial risk protection especially for vulnerable populations.

Greater granularity of equity analyses contributes to identify subgroups of women and children who are being left behind and monitor the impact of efforts to reduce inequalities in order to achieve the health-related SDGs⁶⁹. The household registration for recipients of BISP, 'sehat sahulat' card and 'Ehsaas' programme remain a unique opportunity to map and target the poorest. The health sector can use district ranking by UHC and Multi-Dimensional Poverty indices to prioritize districts for equity focused programming.

Prioritizing Action to Reach the SDG and Related Targets by 2030

Pakistan can only accelerate progress towards ending preventable mortality for mothers, newborns and children or achieve family planning goals by truly following the principles of National Health Vision Strategy. Despite significant progress over the years, the historical rates of reduction in maternal and child deaths and contraceptive prevalence rate are too low to achieve the SDG targets by 2030. With the COVID-19 pandemic and signs of economic recession, the health budgets may shrink or be diverted to containment measures. Keeping RMNCAH+N as a priority should be considered as an investment in future human capital and building a resilient health system.

⁶⁹ Victora C, Boerma T, Requejo J, et al. Analyses of inequalities in RMNCH: rising to the challenge of the SDGs. *BMJ Glob Health* 2019;4:e001295. doi:10.1136/bmjgh-2018-001295

A minimum package of MNCH and FP interventions can be immediately rolled out. A simulation was run using the Johns Hopkins Lives Saved Tool (LiST) to model the impact of increasing the effective coverage of this bundle of interventions to 80%, from 2021 to 2025. Two scenarios were run with the priority MNCH package without family planning and the priority MNCH package with family planning as envisaged in Government targets.

The bundle of interventions for increasing coverage to 80% in scenario 1 included ANC 4+, skilled attendance at birth, basic and comprehensive Emergency Obstetric Care, essential newborn care, small and sick newborn care, treatment of diarrhoea with ORS + Zinc, treatment of Pneumonia with antibiotics, full immunization, hand washing with soap and improved water supply. In scenario 2, the CPR was increased by 20% in addition to all the interventions in scenario 1.

Table 22: Percent reduction in number of deaths

	Percent reduction in number of deaths		
	Scenario 1: MNCH	Scenario 2: MNCH+FP	Accelerator effect of FP in percentage points
Neonatal deaths	43	58	15
Still births	40	57	17
Maternal deaths	41	59	18
Under-five deaths	33	48	15

The highest number of deaths averted among mothers was related to PPH, eclampsia and sepsis. Among the newborns, maximum mortality reduction was achieved for prematurity and asphyxia. The post-neonatal deaths of children under-five registered the largest decrease for diarrhoea and pneumonia-related deaths. This is understandable, considering the total burden of deaths by cause in these age groups. Apart from the high impact evidence-based interventions for maternal, newborn and child health; it is important to acknowledge the accelerator effect of family planning. This is not impossible considering 22% women already have an unmet need for FP. The 70% institutional deliveries are due to un-tapped counselling and FP service delivery opportunities if post-partum family planning is pushed to scale. With this strategy, Pakistan will definitely move closer to achieving the SDG 2030 targets.

The ten priorities for action in the next ten years (10X10 agenda) are summarized below:

1. Improving RMNCH leadership and coordination mechanisms by strengthening TWGs at national and provincial levels;
2. Developing RMNCH acceleration package including family planning for immediate scale-up;
3. Providing operational support to RMNCH acceleration package by UHC structures or RMNCH+N directorates with quality of care units to provide support for implementing evidence-based MNCH and FP standards;

4. Harnessing the role of a strong private sector, already catering to two-thirds of the RMNCH needs. Partnerships with professional associations and health care commissions can provide leverage in this regard;
5. Targeting the poor, the rural and the un-educated is imperative for reducing the coverage disparities. Linking health sector efforts with poverty reduction, nutrition sensitive and social protection initiatives will bring efficiency and reduce duplication;
6. Resolving the serious health work force crises, nurse midwife ratio to population and attaining sufficient numbers of neonatologists and neonatal nurses;
7. Improving timely referral of complicated cases by expanding pooled ambulance services and linking district level facilities with pre-identified teaching hospitals. Laying out clear referral protocols and responsibilities at each level of health system is needed;
8. Digital innovations for improving data recording and reporting as well as improving LMIS and increasing awareness through mobile messaging;
9. Investing in research and information system for tracking effective quality coverage at least on annual basis; accounting for the private sector; bridging data gaps for CRVS (Civil Registration and Vital Statistics); birth weight measurement, reporting of still births and standardized measures for determining quality of care and cohort tracking. Extensive investment is required to scale-up and streamline MPDSR and link these efforts with continuous quality improvement; and
10. Developing inter-sectoral linkages for addressing the underlying risk factors of low female social status, malnutrition, air pollution and inadequate WASH behaviours.

Pakistan has indeed made some positive strides in improving maternal, newborn and child health of its people in the last decades. Challenges of the health system, and equity and high unmet need for family planning need urgent attention to accelerate progress in the next decade. This calls for fundamental health sector reforms to deliver a high-priority effective RMNCAH package of quality services to the women and children who need it most.



